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Rtf1-dependent transcriptional pausing regulates cardiogenesis

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eLife Assessment

This **important** study shows that a controlled pause in gene reading is required for early heart cells to form during development. The authors demonstrate that loss of this pause prevents the proper activation of the heart-producing program across animal and stem cell systems. The evidence is **compelling**, supported by careful genomic and functional analyses that clearly define the developmental block. Overall, this work will interest developmental biologists and inspire further studies on the origins of early heart defects.

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Abstract

Transcriptional pause-release critically regulates cellular RNA biogenesis, yet how dysregulation of this process impacts embryonic development is not fully understood. Rtf1 is a multifunctional transcription regulatory protein involved in modulating promoter-proximal pausing of RNA polymerase II. Using zebrafish and mouse as model systems, we show that Rtf1 activity is essential for the differentiation of the myocardial lineage from mesoderm. Ablation of *rtf1* impairs the formation of *nkx2.5+tbx5a+* cardiac progenitor cells, resulting in the development of embryos without cardiomyocytes. Structure-function analysis demonstrates that Rtf1's cardiogenic activity requires its Plus3 domain, which confers interaction with the pausing/elongation factor Spt5. In Rtf1-deficient embryos, the occupancy of RNA Pol II at transcription start sites was reduced relative to downstream occupancy, suggesting a reduction in transcriptional pausing. Intriguingly, attenuating pause release by pharmacological inhibition or morpholino targeting of CDK9 improved RNA polymerase II occupancy at the transcription start sites of key cardiac genes and restored cardiomyocytes in Rtf1-deficient embryos. Thus, our findings demonstrate the crucial role that Rtf1-mediated transcriptional pausing plays in controlling the precise spatio-temporal transcription programs that govern early heart development.

Introduction

Cardiogenesis relies on tightly controlled transcriptional programs regulated at all stages of mRNA synthesis. Decades of studies have identified an evolutionarily conserved network of transcription factors that direct embryonic heart development. Notably, members of the GATA, NK-2 homeobox, TBX, HAND, and MEF2 transcription factor families play crucial, conserved, and well-characterized roles in specifying cardiac cells from the mesoderm^{1–4}. Through a careful balance of positive and negative regulatory signals, the cardiac transcription regulatory network coordinates with other transcriptional networks to ensure that the appropriate number of cardiac progenitors are specified. In zebrafish, this activity partitions the anterior lateral plate mesoderm (ALPM) into multiple fates including the cardiac mesoderm, which is bounded anteriorly by blood and endothelial progenitors and posteriorly by pectoral fin progenitors^{5–8}.

Along with the paradigmatic role played by cardiac transcription factors in transcriptional initiation, accumulating evidence suggests that transcriptional control of cardiogenesis is also accomplished by epigenetic and post-initiation mechanisms. For example, the BAF (Brg1/Brm-associated factor) complex, an ATP-dependent chromatin remodeling complex that actively positions nucleosomes, is critical for normal vertebrate cardiogenesis^{9–11}. The BAF complex directly interacts with cardiac transcription factors including Tbx5, Nkx2-5, and Gata4¹² and controls temporal changes in chromatin accessibility during cardiac differentiation⁹. Intriguingly, forced expression of the BAF component Baf60c along with Tbx5/Nkx2-5/Gata4 can induce cardiac differentiation from non-cardiac mesoderm¹³, suggesting that nucleosomal positioning and chromatin accessibility are key mechanisms regulating the activity of cardiac transcription factors during cardiogenesis.

Cardiac gene expression is also sensitive to transcription elongation regulation. Spt6 (Suppressor of Ty 6) is a histone chaperone that promotes RNA Polymerase II elongation and co-transcriptionally modulates chromatin structure^{14,15}. In zebrafish, Spt6 interacts genetically with the transcriptional pausing/elongation factor Spt5 (Suppressor of Ty 5) to support Nkx2.5+ cardiac progenitor specification¹⁶. While both Spt6 and Spt5 promote transcription elongation, they likely do so via distinct mechanisms. Spt5 directly interacts with Spt4 to form the DSIF (DRB Sensitivity Inducing Factor) complex, which has both inhibitory and stimulatory roles in transcription¹⁷. Unphosphorylated Spt5 fosters promoter-proximal pausing of RNA Polymerase II, but phosphorylation of Spt5's C-terminal repeat region by Cdk9 (cyclin dependent kinase 9) promotes transcriptional pause release, and converts DSIF into an elongation-promoting factor^{18–20}. How the transcription regulatory activities of these elongation and pausing factors facilitate cardiac development is not yet understood.

The Polymerase Associated Factor 1 Complex (PAF1C) consists of five core proteins (Paf1, Ctr9, Cdc73, Rtf1, and Leo1) and plays wide-ranging transcription regulatory roles including promoting co-transcriptional epigenetic modification of histones, supporting both pausing and elongation of RNA Polymerase II, influencing mRNA 3' end formation, and serving as a bridge between specific transcription factors and RNA Polymerase II^{21–23}. In zebrafish, PAF1C subunits differentially regulate cardiac development²⁴. Leo1 activity is dispensable for the formation of cardiac progenitors, but is required for differentiation of the cardiac chambers²⁵. Embryos lacking Ctr9, Cdc73, or Paf1, on the other hand, exhibit a deficiency of cardiomyocytes and only form small, dysmorphic hearts. Intriguingly, loss of the Rtf1 subunit of the PAF1C results in the most dramatic defects in cardiogenesis. Rtf1-deficient embryos have severe reduction in *nkx2.5*+ cardiac progenitors and fail to generate any cardiomyocytes²⁴.

Rtf1 is a multifunctional transcription regulatory protein that modulates pausing and elongation of RNA polymerase II, as well as histone epigenetic modifications. Given Rtf1's numerous roles in transcription regulation, it remains an open question how Rtf1 mechanistically regulates cardiogenesis. In this study, we showed that zygotic *rtf1* mutants, like *rtf1* morphants, lack expression of most cardiac progenitor marker genes and produce no cardiomyocytes in zebrafish. We also found that knockout of Rtf1 in the cardiogenic mesoderm of mouse embryos and knockdown of Rtf1 in mouse ESCs reduced the formation of cardiac tissue, indicating that the role of Rtf1 in specification of the cardiac lineage is conserved among vertebrates. Using FACS and transcriptomic profiling approaches, we show that differentiation of the lateral plate mesoderm (LPM) into cardiac progenitors requires Rtf1 activity. We further show that Rtf1's Plus3 domain is essential for cardiac progenitor formation, while its HMD (Histone modification domain) is dispensable. Consistent with the Plus3 domain's function as an interaction point between Rtf1 and the DSIF component Spt5, Rtf1-deficient embryos display a genome-wide decrease in promoter-proximal pausing of RNA Polymerase II. Excitingly, attenuating pause release by chemical and morpholino inhibition of CDK9 improved pausing at cardiac genes and restored cardiac progenitor formation to Rtf1-deficient embryos. Together, our findings suggest that during development, Rtf1 facilitates expression of the cardiac transcription program by supporting the promoter-proximal pausing of RNA Polymerase II.

Results

Rtf1 is essential for heart development in zebrafish

We previously showed that knockdown of *rtf1* abolishes cardiac progenitor cell formation in zebrafish. To corroborate this finding, we used CRISPR/Cas9 to generate stable *rtf1* mutant lines. We identified two independent lines (*rtf1*^{LA2678} and *rtf1*^{LA2679}) with small deletions in exon 3 of the *rtf1* gene that are predicted to result in frameshifts and early stop codons, which nearly eliminate the production of Rtf1 protein (Fig. 1a-c). Like *rtf1* morphants and the previously reported *rtf1*^{KT641} allele^{24,26}, both *rtf1*^{LA2678} and *rtf1*^{LA2679} mutants have poorly defined posterior somite boundaries but are otherwise visually indistinguishable from their wild type siblings during early somitogenesis, the developmental timeframe when cardiac progenitors emerge from the LPM. After one day of development, cardiac tissue is not detectable in *rtf1* mutant embryos by visual inspection or by *myl7* expression (Fig. 2a,b) and mutant embryos display a downwardly curved body shape and an absence of pigmentation at two days post-fertilization (Fig. 2 Supplement 1). In situ hybridization further showed that *rtf1*^{LA2678} and *rtf1*^{LA2679} mutant embryos, like *rtf1* morphants, fail to express normal levels of critical cardiac transcription factors including *mef2ca*, *nkx2.5*, *tbx5a* and *tbx20* at the time when cardiac progenitors emerge from the ALPM (Fig. 2c-j). Together, these data show that morpholino knockdown faithfully recapitulates *rtf1* mutant phenotype²⁴ and demonstrate that Rtf1 activity is required for the expression of multiple cardiac transcription factors, indicating a critical role of Rtf1 in regulating the cardiac gene program.

Conserved role of Rtf1 in mammalian cardiac development

The sequence of the Rtf1 protein is highly conserved among vertebrates (83% identity at the amino acid level between fish and humans). To examine if the crucial role of Rtf1 in cardiogenesis we observed in zebrafish was conserved in mammals, we expressed Cre-recombinase under the control of the *Mesp1* promoter in Rtf1 flox mouse embryos (Fig. 3a).²⁸ The *Mesp1* promoter drives expression of Cre in the early cardiogenic mesoderm (E6.25),²⁹ allowing for knockout of Rtf1 activity prior to the onset of heart development. Rtf1 *Mesp1*-Cre knockout mouse embryos die prior to birth, suggesting that Rtf1 function in the cardiogenic mesoderm is essential for survival. We examined the developing hearts of Rtf1 *Mesp1*-Cre knockouts and their wild type siblings by in situ hybridization and found that while Cre+;Rtf1 flox/+ and Cre-embryos display an *Nkx2.5*/*Tbx20*+ heart tube undergoing looping at E8.5, Rtf1 *Mesp1*-Cre knockouts have little or no expression of these markers (Fig. 3b,c). These data suggest a conservation of the critical role of Rtf1 in differentiation of the myocardial lineage from the mesoderm.

To further investigate the conservation of Rtf1's role in cardiac differentiation, we knocked down Rtf1 activity in mouse embryonic stem cells (mESCs) using short hairpin RNA (Fig. 4a,b). Unmanipulated embryoid bodies (EBs) derived from mESCs are capable of differentiating into cardiac tissues and form beating patches (Fig. 4c,d). In sharp contrast, Rtf1-deficient mESCs display a dramatic reduction in *Myh6*, *Nkx2.5*, and *ANF/Nppa* expression and produce significantly fewer beating patches (~67% reduction) (Fig. 4a-d). Interestingly, the temporally dynamic expression levels of *Brachyury* (mesoderm), *Afp* (endoderm) and *Ncam1* (ectoderm) are not significantly affected by Rtf1 knockdown (Fig. 4e-g). Together, these data show that Rtf1 is dispensable for the formation of the three germ layers but is essential for differentiation of the cardiac lineage from mesoderm in mammals. This phenotype is similar to what we observed in zebrafish embryos, indicating a conservation of Rtf1's function in cardiogenesis between fish and mammals.

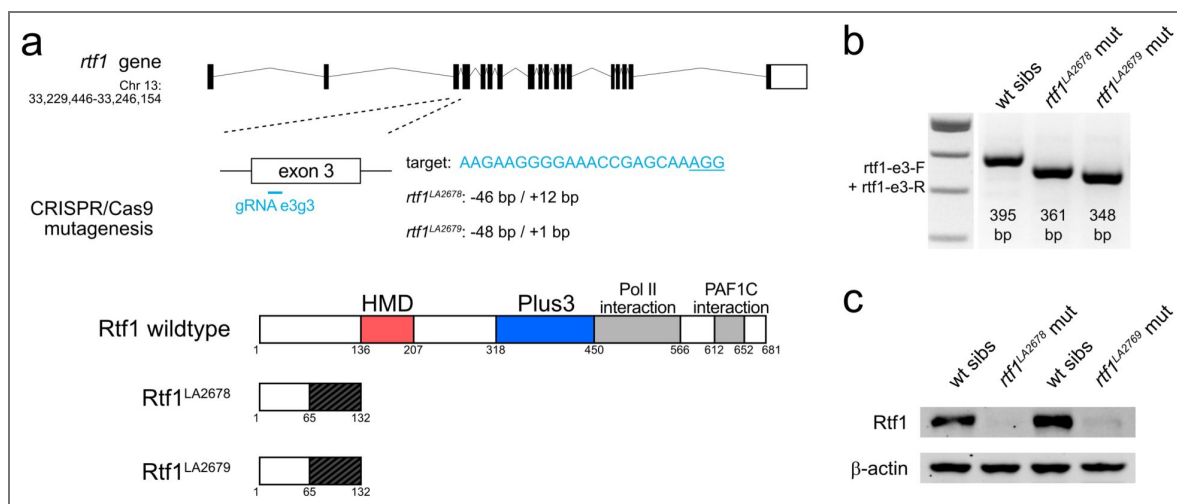


Figure 1. Generation of zebrafish *rtf1* null mutants with CRISPR/Cas9.

(a) Schematic of CRISPR/Cas9 mutagenesis of the *rtf1* gene. Two mutant alleles, *rtf1*^{LA2678} and *rtf1*^{LA2679}, were recovered after targeting of *rtf1* exon 3. Both alleles are predicted to disrupt translation of the Rtf1 protein and eliminate the HMD, Plus3, Pol II interaction, and PAF1C interaction domains. (b) Agarose gel electrophoresis results of genotyping *rtf1* mutants by PCR. Deletions in *rtf1*^{LA2678} and *rtf1*^{LA2679} alleles can be distinguished from wild type allele using primers rtf1-e3-F and rtf1-e3-R. (c) Western blot detecting Rtf1 and β-actin (loading control) proteins in lysates of wild type and *rtf1* mutant embryos. Image is representative of two independent experiments.

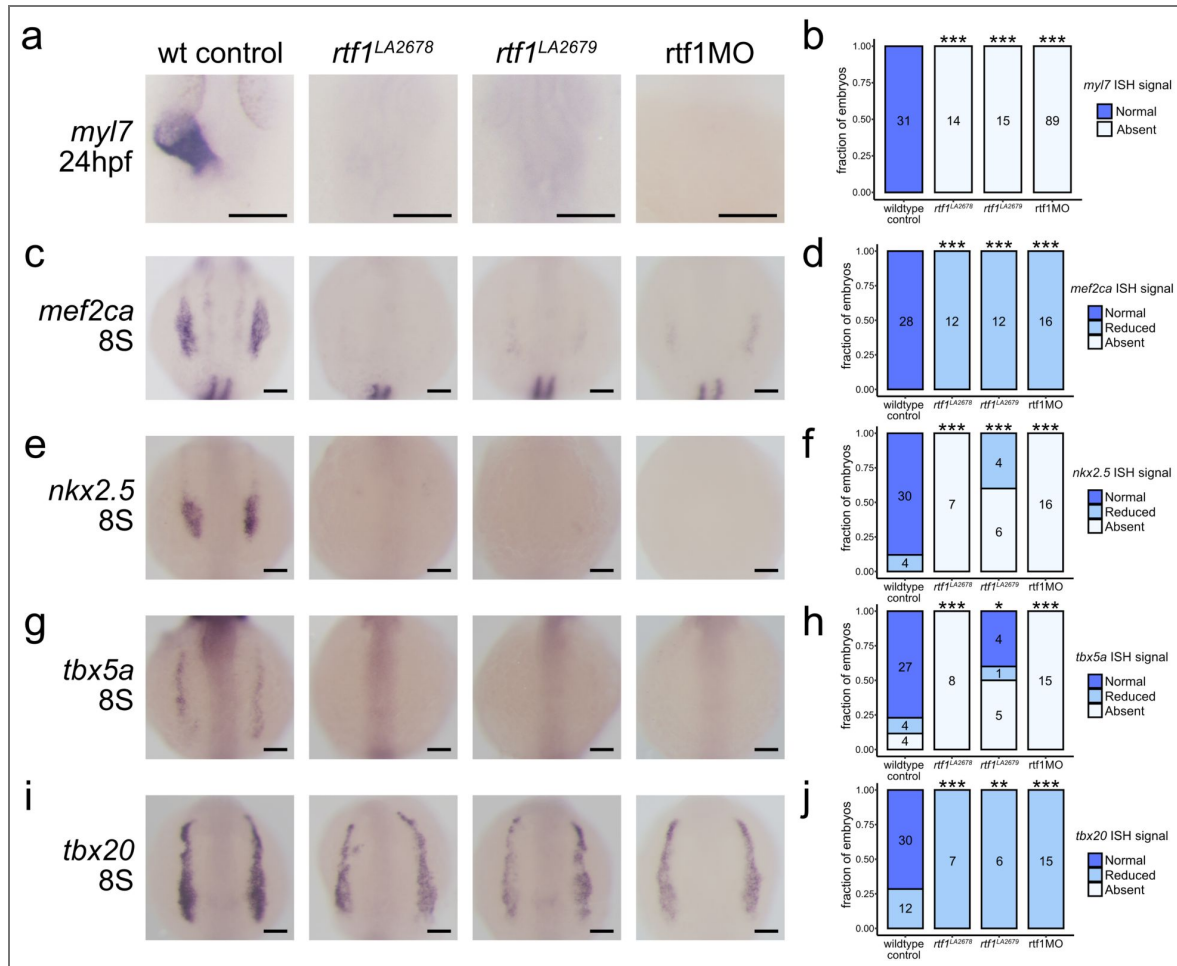


Figure 2. Characterization of early cardiac development in Rtf1-deficient embryos.

(a) Representative images of RNA in situ hybridization detecting *myl7* expression in 24 hpf zebrafish embryos. (b) Quantification of *myl7* signal intensity in control and Rtf1-deficient embryos at 24 hpf. Numbers on bars indicate the number of embryos analyzed. (c) Representative images of RNA in situ hybridization detecting *mef2ca* expression in 8 somite stage zebrafish embryos. (d) Quantification of *mef2ca* signal intensity in control and Rtf1-deficient embryos at the 8 somite stage. Numbers on bars indicate the number of embryos analyzed. (e) Representative images of RNA in situ hybridization detecting *nkx2.5* expression in 8 somite stage zebrafish embryos. (f) Quantification of *nkx2.5* signal intensity in control and Rtf1-deficient embryos at the 8 somite stage. Numbers on bars indicate the number of embryos analyzed. (g) Representative images of RNA in situ hybridization detecting *tbx5a* expression in 8 somite stage zebrafish embryos. (h) Quantification of *tbx5a* signal intensity in control and Rtf1-deficient embryos at the 8 somite stage. Numbers on bars indicate the number of embryos analyzed. (i) Representative images of RNA in situ hybridization detecting *tbx20* expression in 8 somite stage zebrafish embryos. (j) Quantification of *tbx20* signal intensity in control and Rtf1-deficient embryos at the 8 somite stage. Numbers on bars indicate the number of embryos analyzed. Scale bars in a, c, e, g, and i represent 0.1 mm. *: $p < 0.05$, **: $p < 0.01$, ***: $p < 0.001$ based on Fisher's Exact Test.

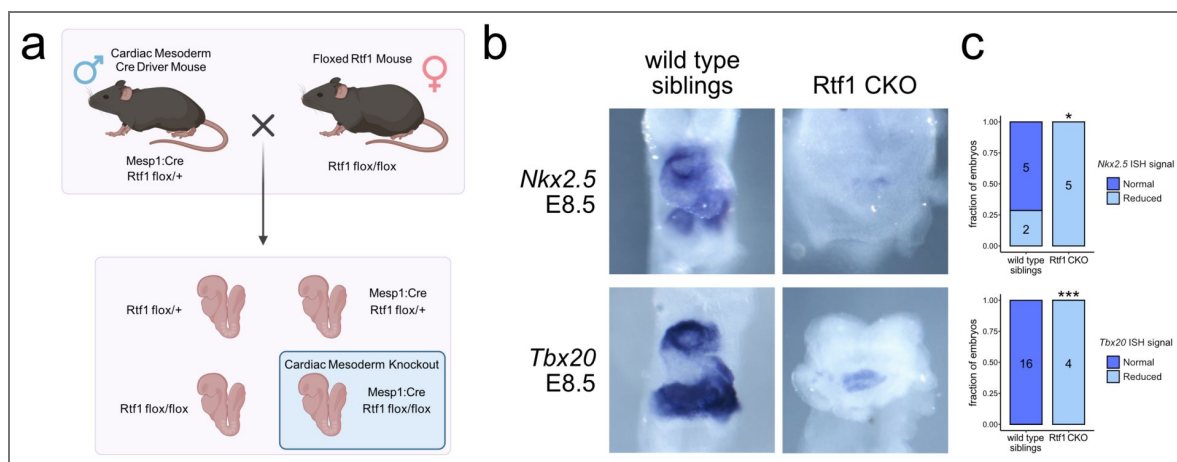


Figure 3. Failure of cardiac differentiation upon loss of Rtf1 in the mammalian cardiac mesoderm.

(a) Diagram of generation of cardiac mesoderm-specific Rtf1 knockout mouse embryos. A Mesp1:Cre insertion allele was bred into an Rtf1 exon 3 floxed background. Heterozygous Rtf1 floxed male mice with Mesp1:Cre were bred to homozygous Rtf1 floxed females, resulting in one quarter of embryos lacking Rtf1 activity in the cardiac mesoderm. Created with [BioRender.com/teaid80](https://www.biorender.com/teaid80). (b) Representative images of RNA in situ hybridization detecting *Nkx2.5* (wild type n = 7, Rtf1 CKO n = 5) and *Tbx20* (wild type n = 16, Rtf1 CKO n = 4) gene expression in Cre-negative (wild type siblings) and Cre-positive homozygous Rtf1 floxed (Rtf1 CKO) embryos at E8.5. (c) Quantification of *Nkx2.5* and *Tbx20* signal intensity in Cre-negative (wild type siblings) and Cre-positive homozygous Rtf1 floxed (Rtf1 CKO) embryos at E8.5. Numbers on bars indicate the number of embryos analyzed. *: $p < 0.05$, ***: $p < 0.001$ based on Fisher's Exact Test.

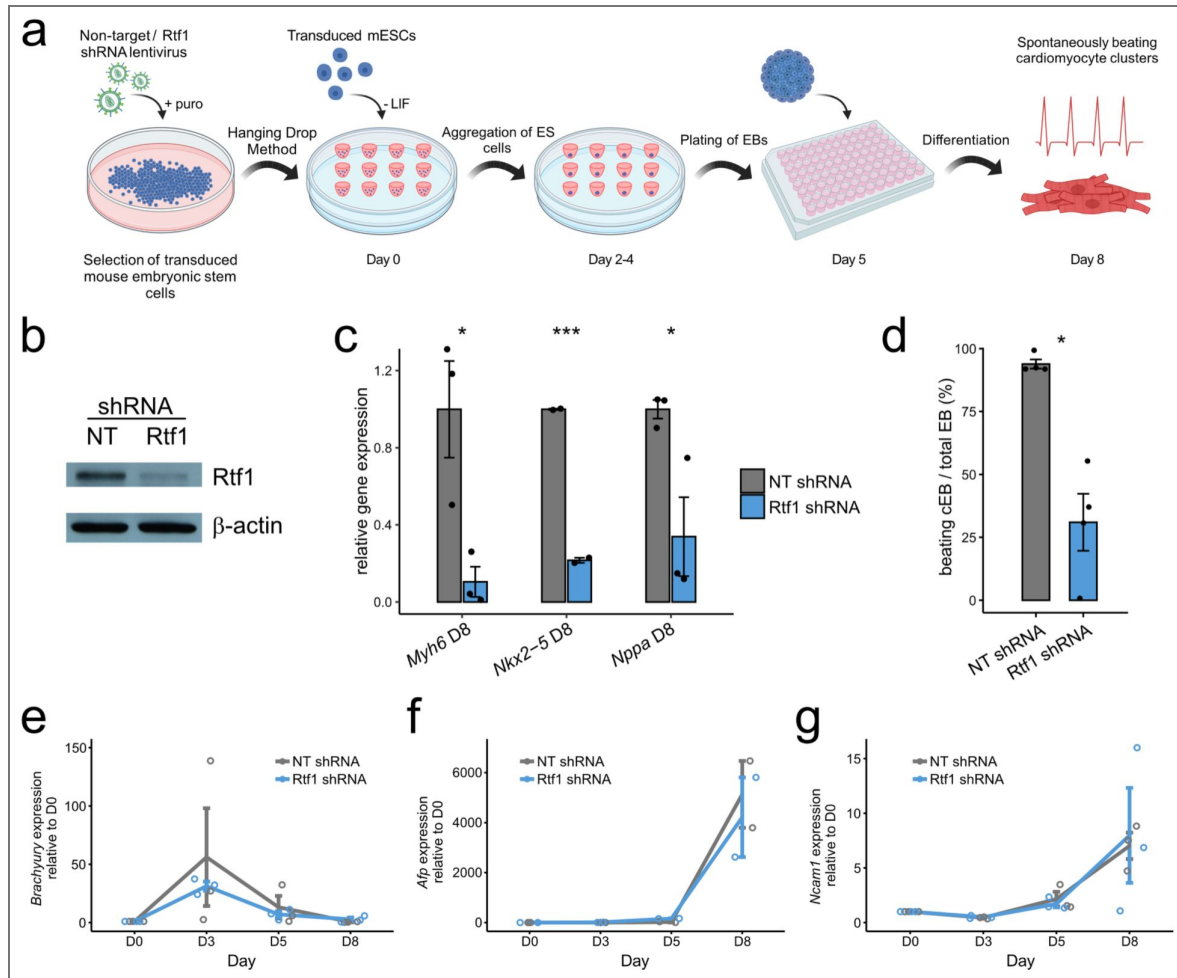


Figure 4. Failure of cardiac differentiation from Rtf1 knockdown mouse embryonic stem cells.

(a) Diagram of shRNA knockdown of Rtf1 in mouse embryonic stem cells (mESCs) and differentiation of plated EBs. mESCs were transduced with non-target control or Rtf1 shRNA lentivirus and selected with puromycin. Transduced mESCs were grown into embryoid bodies (EBs) using the hanging drop method which were then plated to examine differentiation into cell types including beating cardiomyocyte clusters. Created with BioRender.com/k5x2m7j. (b) Western blot verifying reduction of Rtf1 protein in Rtf1 shRNA mESCs (~70% reduced based on densitometry) compared to unchanged level of the loading control protein β-actin. Image is representative of three independent experiments. (c) qPCR analysis of *Myh6*, *Nkx2-5*, and *Nppa* expression in non-target control (NT) and Rtf1 shRNA plated EBs. Data are normalized to the mean expression level in NT shRNA samples and error bars indicate the standard error of the mean. (d) Mean percentage (± standard error) of plated EBs exhibiting beating cardiomyocyte differentiation (cEBs) in NT and Rtf1 shRNA plated EBs. (e) qPCR analysis of *Brachyury* expression in non-target control (NT) and Rtf1 shRNA plated EBs. (f) qPCR analysis of *Afp* expression in non-target control (NT) and Rtf1 shRNA plated EBs. (g) qPCR analysis of *Ncam1* expression in non-target control (NT) and Rtf1 shRNA plated EBs. Data in e, f, and g are normalized to the mean expression level in NT shRNA samples at day 0 (D0) and error bars indicate the standard error of the mean. *: $p < 0.05$, ***: $p < 0.001$.

Rtf1 is required for expression of genes associated with differentiation of cardiac precursors from the lateral plate mesoderm

To gain insight into how Rtf1 deficiency impacts differentiation of the cardiac lineage, we profiled LPM transcriptomes of control and *rtf1* morphant embryos. We previously showed that loss of Rtf1 activity does not disrupt expression of the LPM marker *hand2*²⁴. We therefore took advantage of a *hand2*:GFP zebrafish transgenic reporter line to isolate LPM cells from wild type and Rtf1-deficient embryos (Fig. 5a-c and Fig. 5 Supplement 1). Compared to wild type control samples, the *rtf1* morphant LPM displayed 1420 significantly downregulated and 1185 significantly upregulated genes (adjusted *p*-value < 0.05) at the 10-12 somite stage when cardiac progenitor formation is underway in the LPM (Supplementary Data File 1). We examined the most significantly differentially expressed genes and observed that many of the significantly downregulated genes, including *wnt2bb*, *mef2cb*, *mef2ca*, *isl2b*, *tnnt2a*, *atp2a2a*, *rbfox1l*, and *nkx2.5*, are cardiac-lineage markers or have critical known roles in cardiac development (Fig. 5d). We further found that the most significantly downregulated genes were highly enriched for gene ontologies related to cardiac development, embryonic heart morphogenesis, and muscle cell development (Fig. 5e). These data show that Rtf1 activity is required for expression of a broad cardiac gene program in the LPM that is likely under the control of multiple cardiac transcription factors.

To further analyze the deployment of the cardiac gene program during cardiac progenitor formation, we performed Single Cell Multiome sequencing of control and *rtf1* morphant embryos. Clustering of control and *rtf1* morphant cells based on gene expression and chromatin accessibility resolved 39 cell types corresponding to expected major cell lineages (Fig. 6 Supplement 1, Fig. 6 Supplement 2, Supplementary Table 2). Interestingly, while *rtf1* morphant cells of each cell type were present, the frequencies of cell types differed significantly between controls and *rtf1* morphants (Fig. 6 Supplement 3). Notably, LPM cells are significantly more abundant in *rtf1* morphants and ALPM derivative cell types (putative cardiac mesoderm, endothelial precursors and rostral blood precursors) are less abundant, indicating that Rtf1 deficiency negatively impacts ALPM differentiation (Fig. 6 Supplement 3).

We next examined developmental trajectories within the ALPM by reclustering ALPM cells (*sema3e*+/*gata5*+ LPM) and ALPM derivatives (Fig. 6a,b). By calculating pseudotime trajectories, we observed a gradual increase in the expression levels of cardiac markers including *ttn.2*, *mef2cb*, *tnnt2a*, *ryr2b*, and *myh7bb* along the ALPM-cardiac developmental trajectory in control embryos (Fig. 6c-e). In *rtf1* morphants, the proportion of cells representing undifferentiated ALPM was increased (37% in *rtf1* morphants vs 15.5% in controls, Fig. 6b and Fig. 6 Supplement 4). Some cells (26.6%) did appear to differentiate along the cardiac trajectory in *rtf1* morphants; like control cardiac progenitors, they maintained the expression of *gata5*, downregulated expression of the ALPM marker *sema3e*, and initiated expression of the early cardiac lineage genes *bmp6*, *ttn.2*, and *mef2cb* (Fig. 6d-g, Fig. 6 Supplement 4, Fig. 6 Supplement 5). However, loss of Rtf1 activity prevented the proper expression of cardiac lineage-restricted markers that are normally activated later in the ALPM-cardiac trajectory including *tnnt2a*, *ryr2b*, and *myh7bb* (Fig. 6d-g, Fig. 6 Supplement 4, Fig. 6 Supplement 5). Taken together, these data suggest that Rtf1 activity is required for gene expression changes associated with proper cardiac progenitor differentiation.

Rtf1's Plus3 functional domain is necessary for cardiac progenitor formation

The Rtf1 protein is a multifunctional platform for transcription regulation: it consists of several defined functional domains that mediate its interaction with other transcription regulatory proteins and serve as additional points of contact for these proteins with the transcription apparatus (Fig. 1a and Fig. 7a)^{30,31}. The N-terminal half of Rtf1 notably contains its Histone Modification Domain (HMD), which mediates its interaction with E2 ubiquitin-conjugating enzymes and supports the ubiquitylation of histone H2B K120³²⁻³⁴. Rtf1's C-terminus contains the

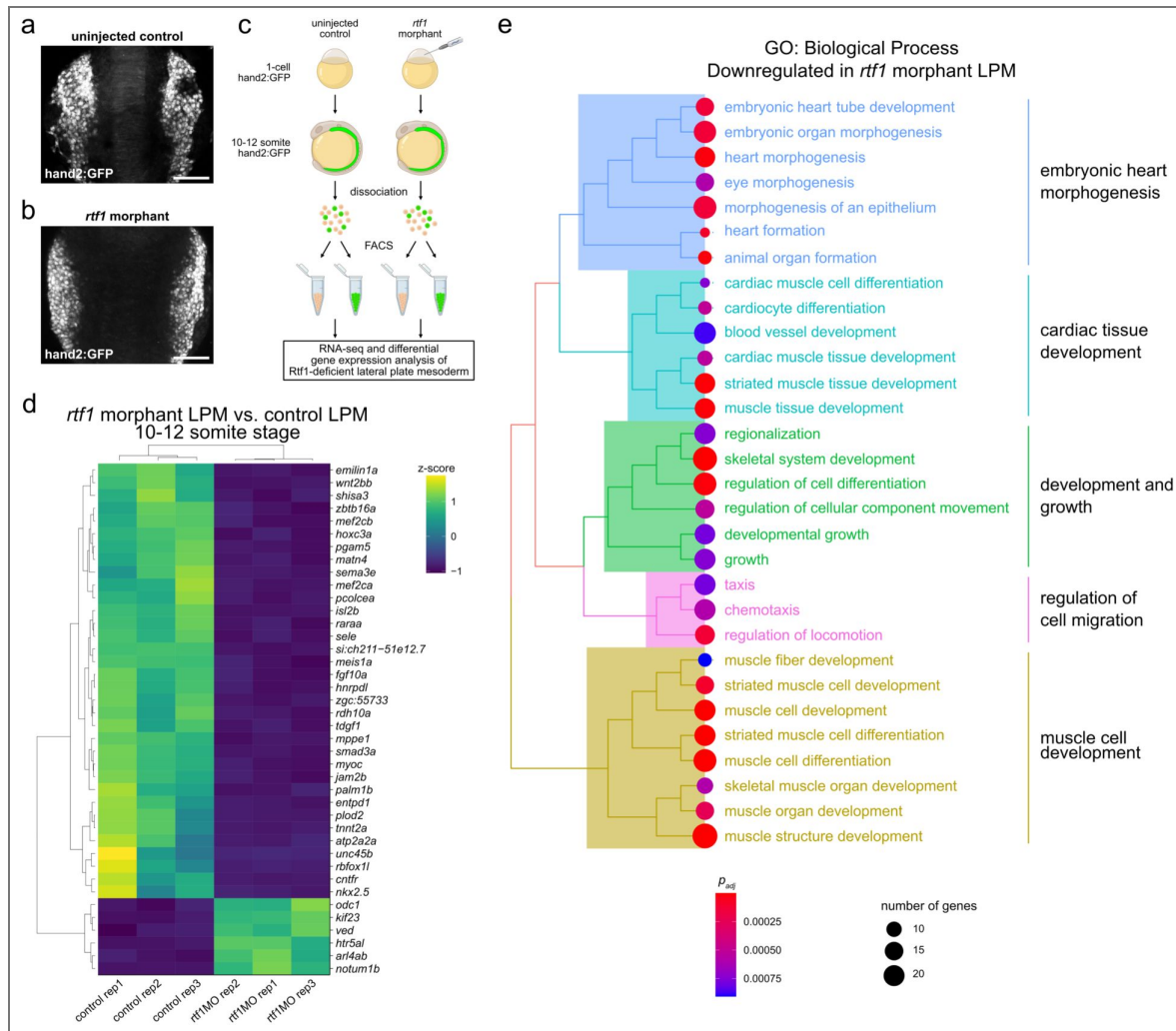


Figure 5. Dysregulation of cardiac gene expression in the *Rtf1*-deficient lateral plate mesoderm.

(a,b) Projections of confocal z-stacks of hand2:GFP signal in 11 somite stage uninjected control (a) and *rtf1* morphant (b) embryos. Scale bars represent 100 μ m. (c) Diagram of lateral plate mesoderm (LPM) FACS RNA-seq experiment. Transgenic hand2:GFP embryos were injected at the 1-cell stage with *rtf1* morpholino and grown to the 10-12 somite stage prior to dissociation and sorting based on GFP expression. RNA isolated from GFP-positive LPM cells was subjected to RNA-seq and compared to RNA from uninjected embryo LPM cells. Created with BioRender.com/0hs986e. (d) Heatmap of scaled expression levels of the top 40 genes most significantly differentially expressed (based on *p*-value) between uninjected control and *rtf1* morphant hand2:GFP-positive LPM. High and low z-scores represent high and low gene expression, respectively. (e) Hierarchical clustering of Biological Process gene ontology (GO) terms based on semantic similarity for the genes most significantly downregulated in the *rtf1* morphant hand2:GFP-positive LPM.

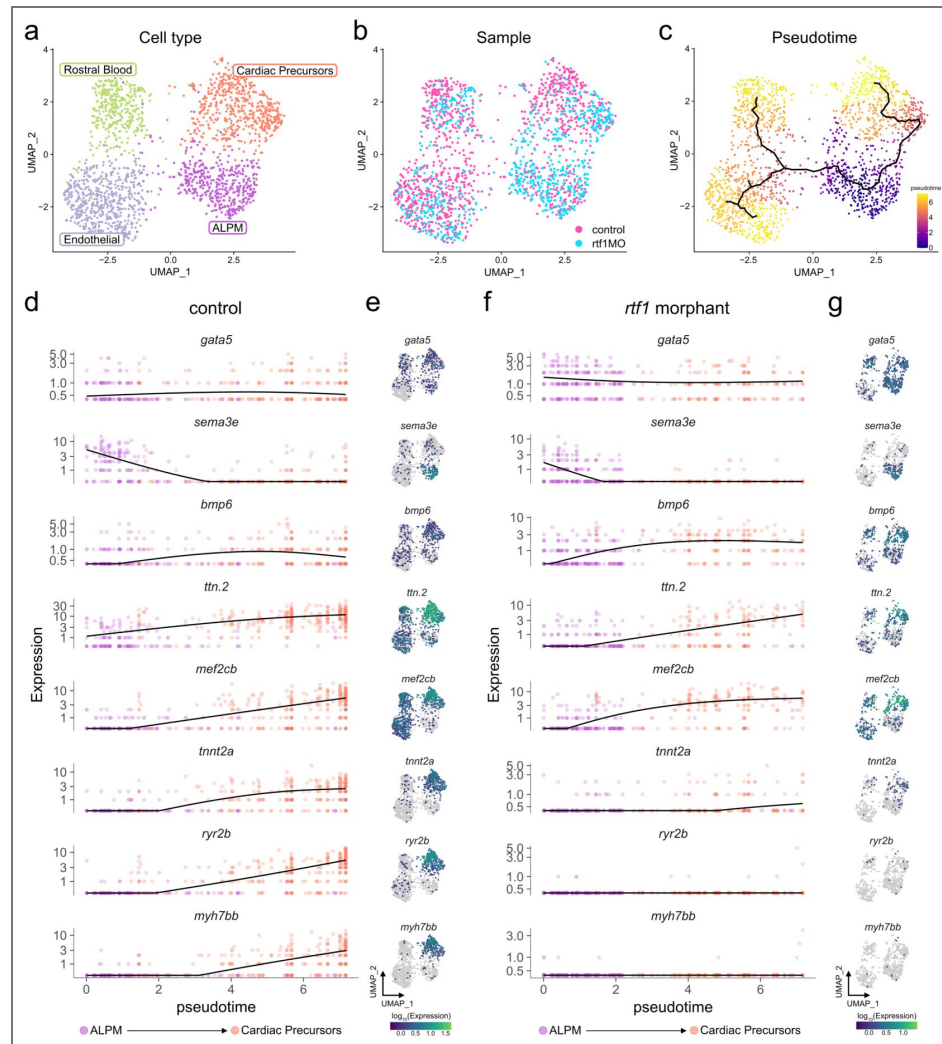


Figure 6. Rtf1 is required for transcriptional progression of cardiac precursors.

(a-c) UMAP plots of anterior lateral plate mesoderm (ALPM) and derivatives from merged and integrated single cell RNA-seq datasets of 11-12 somite stage uninjected control and *rtf1* morphant embryos. Cells in plots are colored by cell type (a), sample (b), and pseudotime (c). (d,f) Expression dynamics plots displaying expression levels (y-axis) for selected genes in uninjected control (d) and *rtf1* morphant (f) cells over the pseudotime trajectory (x-axis) from ALPM (root) to cardiac precursor state shown in (c). (e,g) UMAP plots of uninjected control (e) and *rtf1* morphant (g) cells colored by gene expression levels.

Plus3 domain, which interacts with the C-terminal repeat region of the pausing and elongation factor Spt5^{18,19,35}, a Pol II-interaction domain that allosterically stimulates transcription elongation³¹, and a PAF1C interaction domain.

We investigated if Rtf1 could function in a modular fashion, and if a specific domain or domains were required for the formation of cardiac progenitors during development. To this end, we generated several constructs to express Flag-tagged wild type and mutant versions of Rtf1 lacking specific functional domains and tested whether they could support cardiogenesis in an *rtf1* morphant background. These constructs also carry 8 silent mutations that alter the *rtf1* translation blocking morpholino binding site to ensure successful translation of synthesized mRNA. Wild type (Rtf1 wt*) and mutant FLAG-tagged Rtf1 proteins lacking the HMD (Δ HMD) or Plus3 domain (Δ Plus3) were expressed at comparable levels and properly localized to the nucleus demonstrating that the activity of these proteins was not compromised by instability or mislocalization (Fig. 7b). Injection of Rtf1 wt* mRNA into *rtf1* morphant embryos was able to robustly rescue the knockdown phenotype and support normal formation of the embryonic heart tube (Fig. 7c). Interestingly, we found that Rtf1's HMD was dispensable for its role in cardiac progenitor formation (100% of Rtf1 wt* injected *rtf1* morphants with *nkx2.5* signal, n=25 vs. 100% of Rtf1 Δ HMD *rtf1* morphants with *nkx2.5* signal, n=25) (Fig. 7c). On the other hand, Rtf1's Plus3 domain was essential for supporting the formation of cardiac progenitors (0% of Rtf1 Δ Plus3 *rtf1* morphants with *nkx2.5* signal, n=25) (Fig. 7c). These data suggest that Rtf1's domains function in a modular fashion for regulating cardiac gene expression and point to potential specific effects on transcriptional pausing and elongation.

Rtf1 deficiency disrupts promoter-proximal transcriptional pausing

To examine the occupancy of RNA Pol II, we performed ChIP-seq on embryos at the 10-12 somite stage, when cardiac progenitors are arising from the lateral plate mesoderm. We compared the occupancy of Pol II between uninjected control and *rtf1* morphant embryos and found that Pol II was substantially diminished at the transcription start site (TSS) of many genes compared to the remainder of the gene body (Fig. 8 Supplement 1 and data not shown). Pausing of RNA Pol II at the TSS is a normal aspect of transcription in vertebrates and has been hypothesized to regulate both the timing and expression level of genes^{36,37}. To quantify transcriptional pausing at each gene, we calculated the Pause Release Ratio (PRR), a ratio of Pol II occupancy in the promoter proximal region of a gene (-300 to +300) to the occupancy in the downstream region (+301 to +1301). A higher PRR indicates increased pausing of Pol II near the transcription start site (TSS), while a lower PRR reflects a gene in a less paused state. We found that on a genome-wide scale, pausing is diminished upon loss of Rtf1 activity during embryonic development (Fig. 8a,b). Out of 6,078 genes with substantial Pol II ChIP signal that were analyzed, 5,799 displayed a significantly decreased PRR (Benjamini-Hochberg false discovery rate of 10%) in Rtf1-deficient embryos compared to 37 genes that displayed a significantly increased PRR (Fig. 8b). The median PRR in control embryos was 3.34, while 95% (5,793/6,078) of genes had a PRR of less than 3.34 in Rtf1-deficient embryos (Figure 8a), reflecting a widespread decrease in promoter proximally paused Pol II. These changes in PRR suggest that Rtf1 is critical for normal levels of pausing at transcriptionally active genes during the developmental window when cardiac progenitor cells arise from the lateral plate mesoderm.

Attenuation of pause release permits cardiogenesis in an Rtf1-deficient background

The transition between the pausing and elongation phases of transcription is regulated by PTEF-b, a complex consisting of a cyclin (T1, T2a, or T2b) and the kinase Cdk9³⁸⁻⁴⁰. Phosphorylation of the negative elongation factor (NELF) complex, the pausing/elongation factor Spt5, and the C-terminal tail of RNA Pol II by Cdk9 promotes changes in the factors associated with RNA Pol II and triggers a switch to processive elongation⁴¹. Pharmacological inhibition of Cdk9 with the small molecule flavopiridol can prevent or attenuate pause release, leading to genome-wide increases in pausing

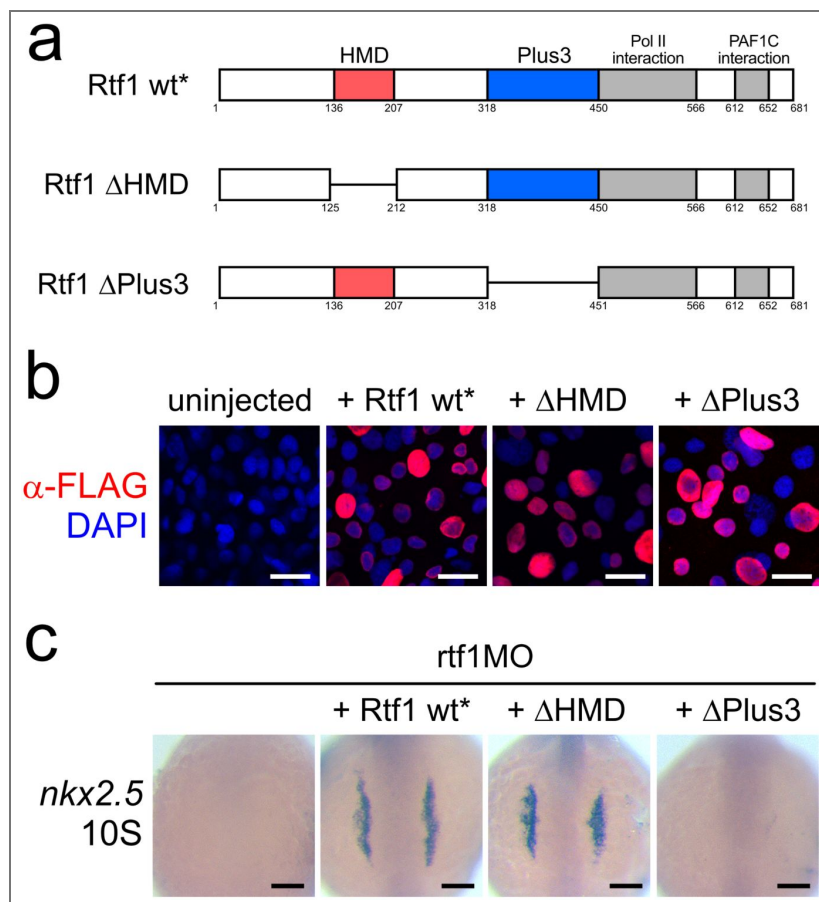


Figure 7. Rtf1's Plus3 domain is necessary for cardiac progenitor formation.

(a) Schematics of Rtf1 wildtype (wt*) and Rtf1 mutant constructs (Δ HMD and Δ Plus3). Domains are indicated by colored boxes (HMD: red, Plus3: blue, PolII and PAF1C interaction: grey). Deleted regions are represented by lines. Numbers refer to the amino acid positions in wildtype Rtf1. Rtf1 wt*, Δ HMD, and Δ Plus3 also harbor 8 silent mutations that alter the *rtf1* translation blocking morpholino binding site. (b) Projections of confocal z-stacks of whole mount immunostaining detecting N-terminal FLAG-tagged Rtf1 constructs (red) expressed in 75% epiboly zebrafish embryos. Nuclei are labeled by DAPI staining (blue). Scale bars represent 20 μ m. (c) Representative images of RNA in situ hybridization detecting *nkx2.5* expression in 10 somite stage (10S) zebrafish embryos coinjected with *rtf1* morpholino and mRNA encoding Rtf1 wildtype (wt*) or mutant (Δ HMD and Δ Plus3) proteins. Scale bars represent 0.1 mm.

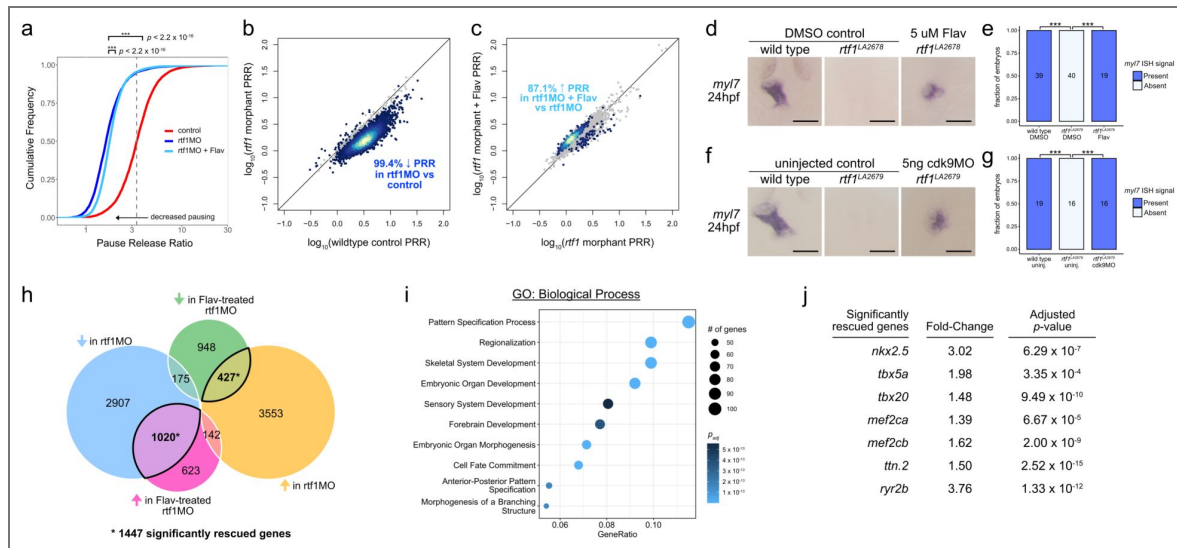


Figure 8. Rtf1-dependent transcriptional pausing regulates cardiogenesis.

(a) Cumulative frequency plot of pause release ratios (PRRs) of 6,078 genes with substantial RNA Pol II signal. Both control (red) and flavopiridol treated *rtf1* morphant (light blue) samples displayed significantly different PRR frequencies compared to untreated *rtf1* morphants (blue). ***: $p < 2.2 \times 10^{-16}$; Welch's paired two-tailed *t*-test. The median PRR of control samples is indicated by a vertical grey dashed line. (b) Dot plot comparing PRRs in controls and *rtf1* morphants. Each dot represents the PRR values for a single gene that differ significantly (colored point) or are not significantly different (grey point) between controls and *rtf1* morphants. (c) Dot plot comparing PRRs in *rtf1* morphants and flavopiridol-treated *rtf1* morphants. Each dot represents the PRR values for a single gene that differ significantly (colored point) or are not significantly different (grey point) between *rtf1* morphants and flavopiridol-treated *rtf1* morphants. Dot colors in (b) and (c) are based on the density of points, with lighter colors indicating more dense points. (d) Representative images of RNA in situ hybridization detecting *myl7* expression in 24 hpf control and *Rtf1*-deficient ($\pm$ flavopiridol) zebrafish embryos. (e) Quantification of *myl7* signal intensity in control and *Rtf1*-deficient ($\pm$ flavopiridol) embryos at 24 hpf. Numbers on bars indicate the number of embryos analyzed. ***: $p < 0.001$. (f) Representative images of RNA in situ hybridization detecting *myl7* expression in 24 hpf control and *Rtf1*-deficient ($\pm$ *cdk9* morpholino) zebrafish embryos. (g) Quantification of *myl7* signal intensity in control and *Rtf1*-deficient ($\pm$ *cdk9* morpholino) embryos at 24 hpf. Numbers on bars indicate the number of embryos analyzed. ***: $p < 0.001$. (h) Venn diagram of significantly altered gene expression in 8-9 somite stage *rtf1* morphant embryos ($\pm$ flavopiridol treatment). Shaded circles represent genes that are significantly: downregulated in *rtf1* morphants vs. uninjected controls (blue), upregulated in *rtf1* morphants vs. uninjected controls (yellow), downregulated in flavopiridol-treated *rtf1* morphants vs. vehicle-treated (DMSO) *rtf1* morphants (green), and upregulated in flavopiridol-treated *rtf1* morphants vs. vehicle-treated *rtf1* morphants (pink). Overlapping regions with thick black outlines represent the 1,447 genes (*) with expression levels that were significantly rescued in 8-9 somite stage *rtf1* morphants by flavopiridol treatment. (i) Plot of enriched Biological Process gene ontology terms for the set of genes that were significantly rescued in *rtf1* morphants by flavopiridol treatment. Dot sizes indicate the number of genes associated with a given ontology term. Dot color indicates the significance level (adjusted *p*-value). (j) Selected list of genes that are critical to cardiac development and were significantly rescued in *rtf1* morphants by flavopiridol treatment. Fold-changes and adjusted *p*-values were calculated with DESeq2. Scale bars in panels d and f represent 0.1 mm.

at most genes⁴². To explore the effect of Cdk9 inhibition on Rtf1-dependent transcription, we performed ChIP-seq for RNA Pol II in flavopiridol-treated *rtf1* morphant embryos at the 10-12 somite stage. Flavopiridol treatment caused a small but highly significant increase in promoter-proximal pausing in an Rtf1-deficient background (Fig. 8a), with 87.1% of genes with significantly altered PRR values displaying an increase in PRR in flavopiridol-treated *rtf1* morphants compared to untreated *rtf1* morphants (Fig. 8c). Furthermore, we found that critical LPM and cardiac genes including *hand2*, *gata5*, *aplnrb*, *bmp4*, *nkx2.7*, and *rbfox1l* exhibited reduced promoter-proximal pausing in Rtf1-deficient embryos that was partially rescued by treatment with flavopiridol (Fig. 8 Supplement 1).

If diminished promoter-proximal pausing at genes crucial for cardiogenesis is responsible for the absence of cardiac progenitor formation in Rtf1-deficient embryos, then we would predict that reducing pause release may restore cardiogenesis in an Rtf1-deficient background. Indeed, flavopiridol treatment (5 µg/mL dosage) potentially restored cardiomyocyte formation in *rtf1* mutant embryos as indicated by *myl7* expression (Fig. 8d,e). Similarly, knockdown of Cdk9 by injection of antisense morpholino oligonucleotides^{43,44} also restored the formation of *myl7*+ cardiomyocytes in an Rtf1-deficient background (Fig. 8f,g), confirming that decreased activity of the Cdk9 kinase is responsible for rescuing the *rtf1* cardiac phenotype. These dramatic results suggest that the primary role of Rtf1 in cardiac progenitor formation is regulating promoter-proximal pausing of critical members of the cardiac gene program.

We next performed RNA-seq to examine what transcriptional changes might account for the rescue of *myl7*+ cardiomyocyte formation in flavopiridol-treated Rtf1-deficient embryos. Strikingly, we found that of the 4,102 genes that were significantly downregulated in *rtf1* morphants at the 8-9 somite stage compared to stage-matched uninjected controls, the expression of 1,020 genes was significantly rescued by treatment with flavopiridol (24.9%; Fig. 8h). Interestingly, we also observed a modest rescue effect on those genes that were significantly downregulated in *rtf1* morphants (427 of 4,122 genes significantly rescued: 10.4%; Fig. 8h). Those genes rescued by flavopiridol treatment were relevant to embryonic development based on significant enrichment for gene ontologies including pattern specification, embryonic organ development, embryonic organ morphogenesis, and cell fate commitment (Fig. 8i). We examined the expression levels of individual genes critical to cardiac development and found that *nkx2.5*, *tbx5a*, *tbx20*, *mef2ca*, *mef2cb*, *ttn.2*, and *ryr2b* expression levels were all significantly rescued by treatment of Rtf1-deficient embryos with flavopiridol (Fig. 8j). Altogether, our data support a model in which loss of Rtf1 causes excessive pause release, a dysregulation of the cardiac transcription program, and a failure in the formation of cardiac progenitors.

Discussion

In this study we showed that the PAF1C member Rtf1 plays a critical role in cardiogenesis in zebrafish and mammals. Ablation of Rtf1 activity impedes mesoderm differentiation along the myocardial trajectory resulting in a failure to produce cardiac progenitors with proper expression of cardiac lineage-restricted genes such as *nkx2.5* and *tbx5a*, and the subsequent loss of the cardiomyocyte population. Structure-function analysis revealed that Rtf1's cardiogenic activity requires its Plus3 domain, a site of interaction with the pausing/elongation factor Spt5, and that loss of Rtf1 diminishes promoter-proximal pausing of RNA Polymerase II. These observations, together with the finding that partial inhibition of transcriptional pause release can support cardiogenesis and restore cardiomyocyte formation in an Rtf1-deficient background, demonstrate a crucial role for Rtf1-mediated transcriptional pausing in early heart development. These findings also raise an intriguing possibility of potentiating cardiac differentiation through manipulating transcriptional pausing.

How exactly pausing regulates gene expression is not fully understood and likely involves both general and gene/locus-specific mechanisms^{36,45}. By limiting the release of RNA Polymerase II into the elongation phase of transcription, pausing can restrain gene expression. This inhibitory role of pausing on transcription allows for another layer of transcription regulation and has been shown to modulate the expression of heat shock and immune stimulus-induced genes^{46–48}. Pausing-

dependent restraint of gene expression is also important for the timing of gene expression in a developmental context. Supporting this notion, Spt5 and NELF-deficient zebrafish embryos exhibit aberrant upregulation of TGF β signaling and a resulting inhibition of hematopoiesis⁴⁴. Intriguingly, the PAF1C has also been shown in mammalian cells to restrain transcription from a subset of promoters by inhibiting nearby enhancer activation⁴⁹. On the other hand, pausing also positively regulates gene expression in some contexts. At genes where nucleosome formation is favored in promoter regions, pausing of RNA Polymerase II can promote a chromatin architecture that is permissive to transcription by competing with nucleosomes for promoter occupancy^{50,51}. This mechanism has been suggested to support expression of the IFN-gamma pathway during zebrafish development⁴⁴. Pausing may also positively regulate gene expression during neural crest development. Loss of Paf1 activity disrupts neural crest differentiation, but this defect can be partially rescued by inactivation of Cdk9. Interestingly, zygotic loss of Cdk9 alone is sufficient to expand the neural crest population in zebrafish embryos, suggesting that promoter-proximal pausing promotes the neural crest gene program⁵².

Experiments performed using a rapid protein degradation system in human cell culture has shed light on the mechanisms by which the PAF1C regulates transcriptional pausing and gene expression outputs⁵³. In DLD-1 cells, acute depletion of PAF1 destabilizes the PAF1C and stimulates pause release. Loss of PAF1 also decreases chromatin recruitment of INTS11, a component of the Integrator-PP2A complex (INTAC) which interacts with PAF1C and counteracts the activity of P-TEFb. This disruption in the balance of INTAC and P-TEFb activities at transcribed genes results in hyperphosphorylation of SPT5 and likely explains the increase in pause release caused by PAF1 depletion. Similar to our finding that Rtf1-deficiency positively and negatively regulates gene expression, depletion of PAF1 resulted in both upregulation and downregulation of many genes. Intriguingly, the gene expression changes caused by PAF1 depletion correlated with the release frequency and processivity of RNA Polymerase II at a given gene, with genes exhibiting low levels of release frequency tending to display decreases in their expression level. These observations suggest that the combined regulatory functions of pausing and elongation determine the transcriptional output of PAF1C-regulated genes. Future studies focused on the chromatin architecture of cardiac gene promoters and other regulatory regions, and the dynamics of pause release and elongation at cardiac genes are needed to provide a complete picture of the manner by which promoter-proximal pausing fosters their expression.

Materials and Methods

Zebrafish

Zebrafish colonies were cared for and bred under standard conditions⁵⁴. Developmental stages of zebrafish embryos were determined based on somite number or hours of development post fertilization at 28.5 °C⁵⁴. Fish husbandry and experiments were performed according to the Institutional Approval for Appropriate Care and Use of Laboratory Animals by the UCLA Institutional Animal Care and Use Committee (Protocols ARC-2000-051-TR-001 and BUA-2018-195-002-CR).

CRISPR mutagenesis

Template DNA for *rtf1* exon 3 guideRNA (rtf1-e3 gRNA) synthesis was amplified with KOD polymerase (MilliporeSigma) using the pXTW-gRNA plasmid and the primers rtf1-e3-F (5'-TAATACGACTCACTATAGGAAGAAGGGGAAACCGAGCAAGTTTGTAGAGCTAGAAATAGC-3') and gRNA-R (5'-AAAAAAGCACCAGCTCGGTGCCAC-3'). rtf1-e3 gRNA (target sequence: 5'-GGAAGAAGGGGAAACCGAGCAA-3') was synthesized using the MEGAshortscript T7 Transcription Kit (ThermoFisher Scientific) and purified using the RNeasy Mini Kit (Qiagen). Zebrafish embryos were injected at the 1-cell stage with 300 pg of Cas9 mRNA and 200 pg of rtf1-e3 gRNA. Injected embryos were raised to adulthood and then outcrossed to wild type AB fish to establish lines carrying specific *rtf1* mutations.

Mice

All mice were maintained in the C57BL/6 background according to the Guide for the Care and Use of Laboratory Animals published by the US National Institute of Health. Housing and experiments were performed according to the Institutional Approval for Appropriate Care and Use of Laboratory Animals by the UCLA Institutional Animal Care and Use Committee (Protocols ARC-2020-045-TR-001 and ARC-2020-043-TR-001). ES cells were obtained from KOMP and used to rederive the *Rtf1*^{tm1a(KOMP)Wtsi} mouse line. This line was then crossed to a ROSA26::FLPe strain⁵⁵ (The Jackson Laboratory, JAX stock #009086) to create an *Rtf1* conditional knockout allele where exon 3 of *Rtf1* is flanked by two loxP sites (*Rtf1* flox). These mice were crossed into the *Mesp1*^{tm2(cre)Ysa} background²⁹ to produce a strain capable of generating *Mesp-Cre*⁺; *Rtf1* flox/flox offspring.

Whole mount in situ hybridization

Whole-mount in situ hybridization was performed as described previously⁵⁶. Embryos for in situ hybridization were fixed in 4% paraformaldehyde in 1xPBS. After a brief proteinase K digestion, embryos were incubated with digoxigenin-labeled antisense RNA probes overnight at 65–70°C. Probes were detected using an anti-digoxigenin antibody conjugated with alkaline phosphatase (Roche 11093274910). The zebrafish antisense RNA probes used in this study include *myl7*, *nkx2.5*, *mef2ca*, *tbx20*, and *tbx5a*. The mouse probes used include *Tbx20* and *Nkx2.5*.

Genotypes of zebrafish embryos were confirmed by PCR with primers:

rtf1-e3-F: 5'-TACAGTGACATCCACCATGCTG-3'

rtf1-e3-R: 5'-GTAAATAGGGGAGAACTAATTGAAG-3'

Genotypes of mouse embryos were confirmed by PCR with primers:

Rtf1-loxP-F: 5'-GACTGAGAAGGGAAATCTTTGAAAC-3'

Rtf1-loxP-R: 5'-CTAGTCCTTGAACAAGTTCAGCT-3'

Cre-F: 5'-GTAGCTGATGATCCGAATAACTAC-3'

Cre-R: 5'-ATCCAGGTTACGGATATAGT-3'.

Constructs

To prevent complementarity between *rtf1* morpholino and mRNA sequences, PCR was used to introduce 8 silent mutations to the morpholino target sequence using pCS2/3XFLAG-*rtf1* as a template²⁴. The morpholino-resistant wild type *rtf1* sequence was cloned into pCS2/3XFLAG to produce an N-terminally FLAG-tagged morpholino-resistant *rtf1* construct (pCS2/3XFLAG-*rtf1*-mr). *Rtf1* deletion mutant sequences were amplified from pCS2/3XFLAG-*rtf1*-mr by SOE-PCR to produce *Rtf1*ΔHMD (missing a.a. 126–211 of *Rtf1*) and *Rtf1*ΔPlus3 (missing a.a. 319–450 of *Rtf1*), which were subsequently cloned into pCS2/3XFLAG to produce N-terminally tagged constructs.

Microinjections

0.5 ng of *rtf1* morpholino (*rtf1*MO; 5'-CTTTCCGTTTCTTTACATTCACCAT-3') was injected at the 1-cell stage^{24,26} along with 1 ng of *p53* morpholino (5'-GCGCCATTGCTTTGCAAGAATTG-3') to prevent *p53*-dependent apoptosis⁵⁷. For *cdk9* knockdown rescue experiments, 5 ng of *cdk9* morpholino (*cdk9*MO; 5'-ACACACAAACATCAAATACTACCC-3')^{43,44} was injected into an incross of *rtf1*^{LA2679} heterozygotes at the 1-cell stage. 150–200 pg of *rtf1* or *rtf1* deletion mutant mRNAs were co-injected with *rtf1* and *p53* morpholinos at the 1-cell stage for mRNA rescue experiments.

Mouse ESC culture and differentiation

E14TG2a mESCs (ATCC) were cultured on 0.1% gelatin covered plates without feeder cells. The feeder-free mESCs were maintained in Dulbecco's modified Eagle's medium (Invitrogen) supplemented with 15% fetal bovine serum (FBS; Invitrogen, USA), 1% non-essential amino acids (Invitrogen), 0.1 mM β-mercaptoethanol (Sigma-Aldrich), 1% (v/v) penicillin-streptomycin-

Glutamine (ThermoFisher Scientific), and 10 $\mu\text{g}/\mu\text{L}$ recombinant mouse leukemia inhibitory factor (LIF; Sigma-Aldrich) in a 5% CO_2 incubator at 37°C. The ESC culture medium was changed every two days. mESCs were subcultured every 3 days using 0.25% trypsin-EDTA (Invitrogen). mESCs were differentiated in hanging drops (600 cells/20 μL) without LIF under non-adherent conditions. Day 5 embryoid bodies (EBs) were plated onto 0.1% gelatin covered plates. Media was changed every two days during adherent culture.

Lentivirus transduction

To knockdown Rtf1 activity in mESCs, Rtf1 or NT (non-target) shRNA lentivirus harboring puromycin resistance were purchased from Sigma-Aldrich. mESCs were transduced with the indicated virus at a multiplicity of infection of 1 and selected in 1.2 $\mu\text{g}/\text{mL}$ puromycin. Rtf1 shRNA knockdown efficacy was tested by Western blotting analysis that detected Rtf1 protein level. Antibodies used were anti-Rtf1 (Bethyl Labs A301-329A) and anti- β -actin (Sigma A1978). Relative protein levels were determined by densitometry measurements using Adobe Photoshop.

Quantitative RT-PCR

Total RNA from NT and shRNA transduced EBs was extracted using TRIzol RNA isolation reagents (ThermoFisher Scientific). cDNA was synthesized using the iScript cDNA Synthesis Kit from Bio-Rad. Real-time quantitative PCR was carried out using the Roche LightCycler 480 Real-Time PCR System. Primers for qPCR are listed in [Supplementary Table 1](#).

Antibody staining

Embryos injected with mRNA encoding FLAG-tagged zebrafish Rtf1 constructs were fixed in 4% PFA in PBS at 75% epiboly. The fixed embryos were incubated in primary antibody (1:50 mouse anti-FLAG M2, F1804, Sigma) in blocking solution (10% goat serum in PBT) for 2 hours at room temperature followed by detection with fluorescent secondary antibody (1:200 anti-mouse IgG-Alexa Fluor 555, A-31570). Nuclei were stained with DAPI and embryos were embedded in 1% low-melt agarose and imaged on a Zeiss LSM800 confocal microscope.

Western Blotting

Embryos were lysed in Rubinfeld's Lysis Buffer at 1 day post fertilization as previously described²⁴. Two embryo equivalents were loaded per lane of an 8% polyacrylamide gel. Following electrophoresis, proteins were transferred to a nitrocellulose membrane and detected with antibodies against Rtf1 (Bethyl Labs A301-329A, 1:4000) and beta-actin (Sigma A1978, 1:5000). ImageJ was used to perform densitometry comparing the levels of Rtf1 protein between samples using beta-actin as the control.

Chemical treatment

Flavopiridol (Cayman Chemicals) was dissolved in DMSO at a stock concentration of 10 mg/mL. Zebrafish embryos were treated with 1.375 – 5 $\mu\text{g}/\text{mL}$ flavopiridol diluted in E3 buffer beginning at 60-70% epiboly. For *rtf1* mutant rescue experiments, genotypes were confirmed using the primers rtf1-e3-F and rtf1-e3-R.

FACS

10-12 somite stage TgBAC(hand2:EGFP)^{pd24} uninjected and *rtf1* morpholino-injected zebrafish embryos⁵⁸ were nutated in 10 mg/mL protease (Protease from *Bacillus licheniformis*, Sigma P5380) in DMEM/F12 media for 30 minutes at 4°C and pipetted to dissociate cells. Debris was removed by filtering through a 40 μm cell strainer and cells were then pelleted by centrifugation at 300 rcf for 5 minutes at 4°C. Cells were resuspended in cold FACS buffer (1xPBS, 2% FBS, 1mM EDTA) and refiltered immediately prior to sorting on a BD FACS Aria II using a 100 μm nozzle. Dead cells were excluded by staining with 0.1 $\mu\text{g}/\text{mL}$ DAPI. Cells from batches of 100 embryos were sorted directly into a mixture of 350 μL Buffer RA1 and 3.5 μL 2-mercaptoethanol from the NucleoSpin RNA kit (Machery-Nagel) for RNA isolation.

mRNA-seq

RNA was purified from FACS-sorted cells with the NucleoSpin RNA kit (Machery Nagel). Total RNA was extracted from whole embryos using TRIzol RNA isolation reagents (ThermoFisher Scientific) and then purified with the NucleoSpin RNA kit (Machery Nagel). Approximately 5–10 ng of input RNA was used to produce libraries with the NEBNext Single Cell/Low Input RNA Library Prep Kit for Illumina (NEB) which were sequenced on an Illumina HiSeq 3000 to produce 50 bp single-end reads. Reads from triplicate samples were mapped to the danRer7 genome using Tophat v2.1.1⁵⁹ with default parameters. Mapped reads were counted with FeatureCounts v2.0.0⁶⁰ and differential expression analysis was carried out with DESeq2 v1.32.0⁶¹. Gene ontology enrichment was examined using the R package clusterProfiler⁶² in R v4.1.0. For gene ontology analysis in LPM FACS experiments, the most significantly downregulated genes in *rtf1* morphant LPM were selected as those genes that were downregulated among the 300 most significantly differentially expressed genes based on adjusted *p*-value. For gene expression analysis of flavopiridol rescue experiments, genes were considered significantly rescued if flavopiridol treatment resulted in a significant expression change in the direction opposite of that caused by *rtf1* knockdown (adjusted *p*-value < 0.1).

ChIP-seq

Embryos for ChIP were dissociated with 4 mg/mL Collagenase Type IV and 0.25% Trypsin at room temperature with intermittent pipetting. Dissociation was stopped with 10% FBS in DMEM and cells were washed once with PBS. Fixation and preparation of chromatin was carried out using the Covaris truChIP Chromatin Shearing Kit and 7 minutes of sonication in a Covaris E220 ultrasonicator using the recommended settings. Immunoprecipitations were carried out as previously described⁶³ on approximately 5 µg of chromatin (minimum 2.5 µg, maximum 8.8 µg) using 5 µL of total RNA Pol II antibody (Cell Signaling D8L4Y). Libraries were prepared using the NEBNext Ultra II DNA Library Kit for Illumina (NEB) and sequenced with an Illumina Novaseq 6000 system to produce 50 bp paired-end reads. Reads from triplicate samples were mapped to the danRer11 genome using bowtie2 v2.4.2⁶⁴ (options: `--local -X 1000`) and read densities of UCSC RefSeq gene regions were quantified using the package csaw v1.24.3^{65,66} in R v4.0.3. For our pausing analyses, we defined the transcription start site (TSS) region as -300 bp to +300 bp surrounding the TSS. RNA Pol II ChIP-seq read densities for the TSS regions of genes displayed a bimodal distribution which was assumed to be two partially overlapping distributions of genes with high and low RNA Pol II TSS signal. We examined those genes with a substantial level of RNA Pol II signal in the TSS region of control samples, defined as those genes present in the upper mode of the distribution of log-transformed TSS read densities based on a cutoff for type-I error of 10% using the R cutoff package v0.1.0 (<https://github.com/choisy/cutoff>). The read density in the TSS region was divided by the read density in the region extending from +301 to +1301 bp with respect to the TSS to calculate the pause release ratio (PRR). ChIP-seq data was visualized in Python 3.10.0 using the plotting tool SparK v2.6.2⁶⁷.

Nuclei isolation for single cell sequencing

Nuclei were isolated from 200 11–12 somite stage zebrafish embryos for 10x Genomics Multiome sequencing using recommended procedures with slight modifications. Embryos were deysolced in Deyolcing buffer (55 mM NaCl, 1.8 mM KCl, 1.25 mM NaHCO₃) and then embryonic cells were partially dissociated by pipetting in DMEM/F12 media (Gibco 21041025). Cells were pelleted and then lysed on ice for 5 minutes in 100 µL of chilled 0.1X Lysis buffer (10 mM Tris pH 7.4, 10 mM NaCl, 3 mM MgCl₂, 0.01% Tween-20, 0.01% IGEPAL CA-630, 0.001% digitonin, 1% BSA, 1 mM DTT). 1 mL of Wash buffer (10 mM Tris pH 7.4, 10 mM NaCl, 3 mM MgCl₂, 0.1% Tween-20, 1% BSA, 1 mM DTT, 0.1 U/µL Sigma Protector RNase Inhibitor) was then added to the cell lysis and the nuclei were pelleted by centrifuging at 500xg for 5 minutes at 4°C. Two more washes with 1 mL of Wash buffer were performed. Nuclei were then resuspended in 50 µL of Nuclei Dilution Buffer (1X 10x Genomics Single Cell Multiome Nuclei Buffer, 1 mM DTT, 1 U/µL Sigma Protector RNase Inhibitor)

and filtered with a 40 μ m Flowmi cell strainer. Approximately 10,000 nuclei per sample were targeted for capture with the 10x Chromium Single Cell Multiome ATAC + Gene Expression system by the UCLA Technology Center for Genomics & Bioinformatics.

Single cell sequencing analysis

10x Chromium Single Cell Multiome ATAC + Gene Expression libraries were prepared by the UCLA Technology Center for Genomics & Bioinformatics following the manufacturer's recommended protocols and sequenced with an Illumina Novaseq 6000 system. Reads from RNA-seq and ATAC-seq assays were aligned to the zebrafish GRCz11 genome with the Lawson laboratory's improved transcriptome annotation⁶⁸ using 10x Genomics' Cell Ranger ARC software v2.0.0 with default parameters. Mapped reads from single cell RNA-seq and ATAC-seq assays were analyzed in R v4.2.2 using Seurat v4.3.0⁶⁹. Good quality cells were retained by filtering using Seurat with the following thresholds: 15000 > RNA UMI > 2000, Genes > 1500, 20000 > ATAC UMI > 5000, TSS enrichment score > 3.5. Filtered control and *rtf1* morphant cell datasets were integrated using Harmony v0.1.1⁷⁰, and preliminary dimensional reduction, cell clustering, and marker gene identification were carried out in Seurat. Clusters of cells that exhibited enriched expression of mitochondrial and ribosomal transcript expression levels but lacked enrichment for other genes were considered abnormal cells and were manually removed from the dataset. Following manual filtering, a final dataset of 14340 control and 13839 *rtf1* morphant cells was again integrated with Harmony and multimodal dimensional reduction (weighted-nearest neighbor method with RNA and ATAC assays), cell clustering, and marker gene identification were carried out in Seurat. Cluster cell number proportions and gene expression levels were visualized with Seurat and with published code⁷¹. Based on marker gene expression, cells that were members of clusters corresponding to anterior lateral plate mesoderm (*sema3e*+), endothelium (*flt4*+), cardiac precursors (*mef2cb*+), and rostral blood (*spi1b*+) were extracted, independently integrated with Harmony, and dimensional reduction (RNA assay only), cell clustering, and marker gene identification were carried out in Seurat. Pseudotime analysis of Harmony/Seurat cell embeddings and plotting of gene expression over pseudotime was performed with Monocle v3.0⁷².

Statistical analysis

Statistical significance for qualitative differences in gene expression in in situ hybridization experiments was determined using Fisher's Exact Test, with a *p*-value of less than 0.05 considered to be significant. Statistical significance of gene expression changes in qPCR analyses was determined based on 2 (for *Nkx2-5* and *Afp* expression) or 3 (for *Myh6*, *Nppa*, *Brachyury*, and *Ncam1* expression) independent experiments using a two-tailed Student's *t*-test with a *p*-value of less than 0.05 considered to be significant. Statistical significance of the difference between beating cardiomyocyte cluster formation in control and *Rtf1* shRNA mESCs was determined based on 4 independent experiments using a two-tailed Student's *t*-test with a *p*-value of less than 0.05 considered to be significant. To compare PRR means between uninjected, *rtf1* morphant, and flavopiridol-treated *rtf1* morphant embryos, mean PRR values were first log-transformed (log10) and then a Welch's paired two-tailed *t*-test was carried out with a *p*-value of less than 0.05 considered to be significant. Significantly changed PRRs for individual genes were determined based on 3 independent replicates per condition using the rowttests function in the R package genefilter v1.72.1 to perform two-tailed *t*-tests. The p.adjust function in the R stats package v4.0.3 using a Benjamini and Hochberg (false discovery rate) correction for multiple comparisons was then applied with adjusted *p*-values of less than 0.1 considered to be significant.

Data availability

The RNA-seq, ChIP-seq, and single cell-seq data presented in this study are available in the NCBI SRA (BioProject PRJNA1015262).



Figure 2 Supplement 1. Bright-field microscopic images of two days post-fertilization (d2) wild type (a) and *rtf1*^{LA2679} mutant (b) embryos.

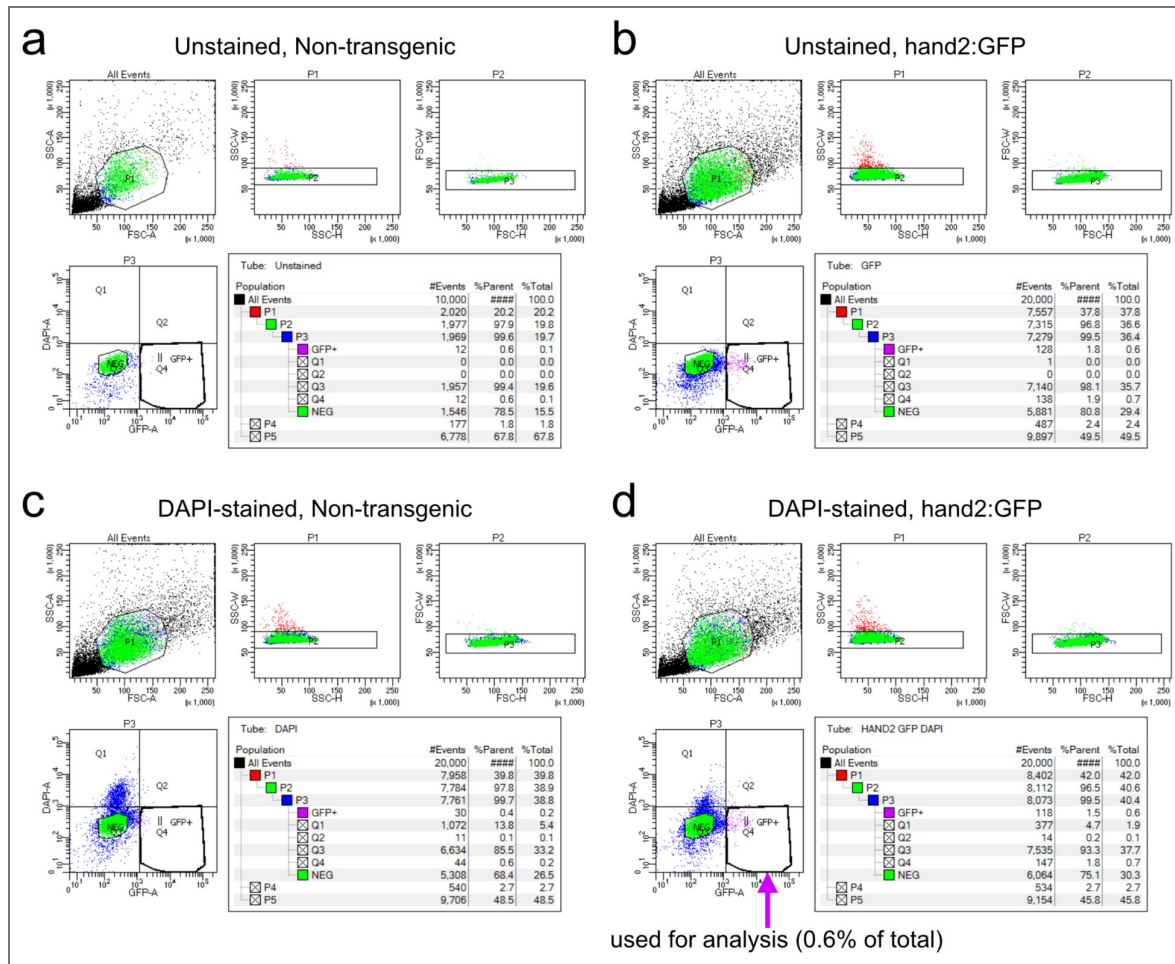


Figure 5 Supplement 1. Gating strategy for 10-12 somite stage zebrafish hand2:GFP FACS.

Abnormal cells or multipleres were excluded by gating out events with high side scatter (SSC) and forward scatter (FSC). Live hand2:GFP-positive cells (pink cells in panel d) were sorted based on high GFP and low DAPI signal. (a) Gating control experiment using unstained cells from non-transgenic embryos. (b) Gating control experiment using unstained cells from hand2:GFP transgenic embryos. (c) Gating control experiment using DAPI-stained cells from non-transgenic embryos. (d) Gating of sorting sample for analysis using DAPI-stained cells from hand2:GFP transgenic embryos. 0.6% of the total cells were sorted as live GFP-positive cells. Plots are representative of 4 independent experiments.

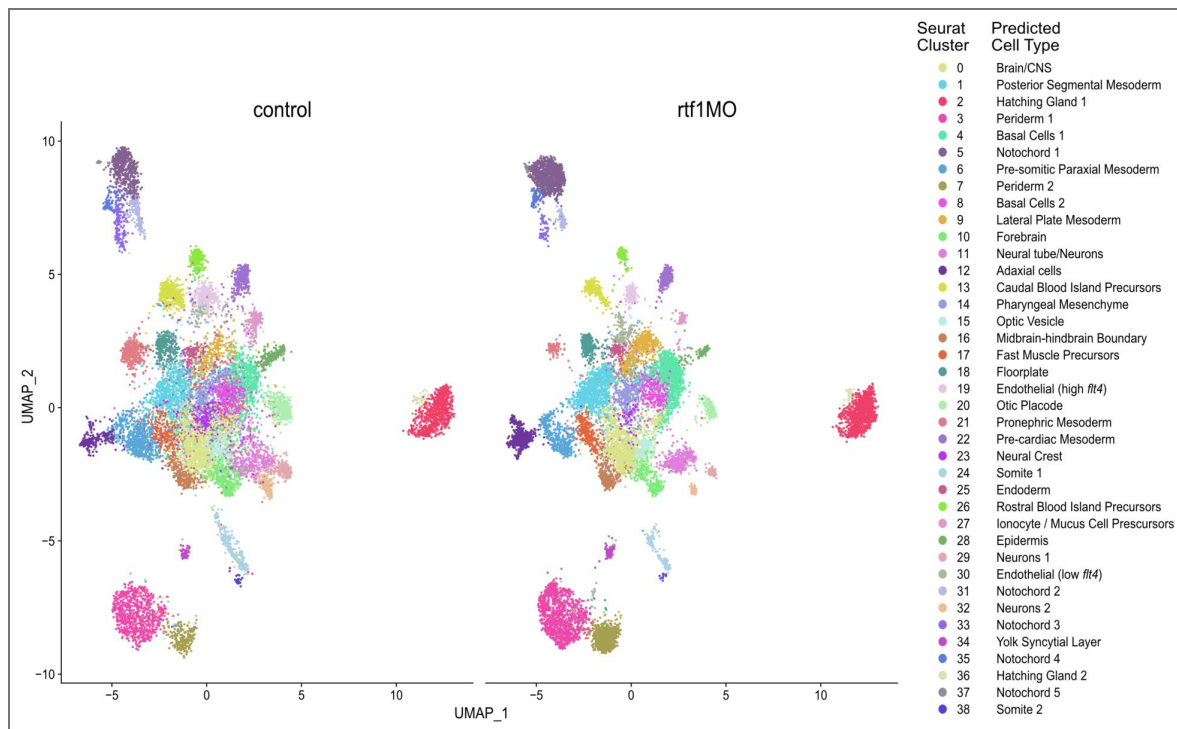


Figure 6 Supplement 1. UMAP plots of 11-12 somite stage control and *rtf1* morphant embryo integrated multimodal (ATAC + Gene Expression) single cell sequencing datasets split based on sample (control on left vs. *rtf1* morphant on right).

Cells are colored based on Seurat cluster. Predicted identities of each cell type are based on manual inspection of marker gene expression.

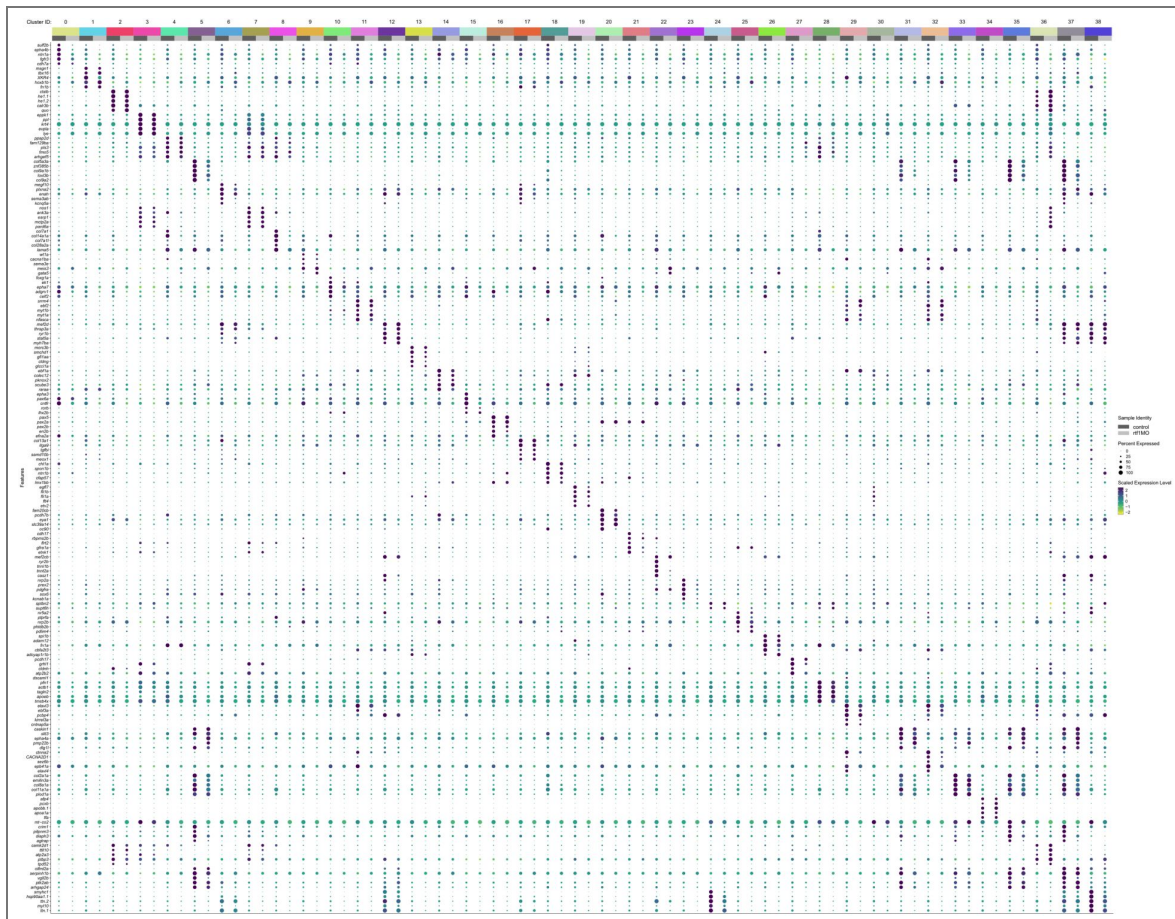


Figure 6 Supplement 2. Clustered dot plot displaying expression of marker genes in cells belonging to each Seurat cluster identified in the integrated multimodal single cell sequencing analysis of 11-12 somite stage control and *rtf1* morphant embryos.

Each cluster is further split by sample (control vs. *rtf1* morphant).

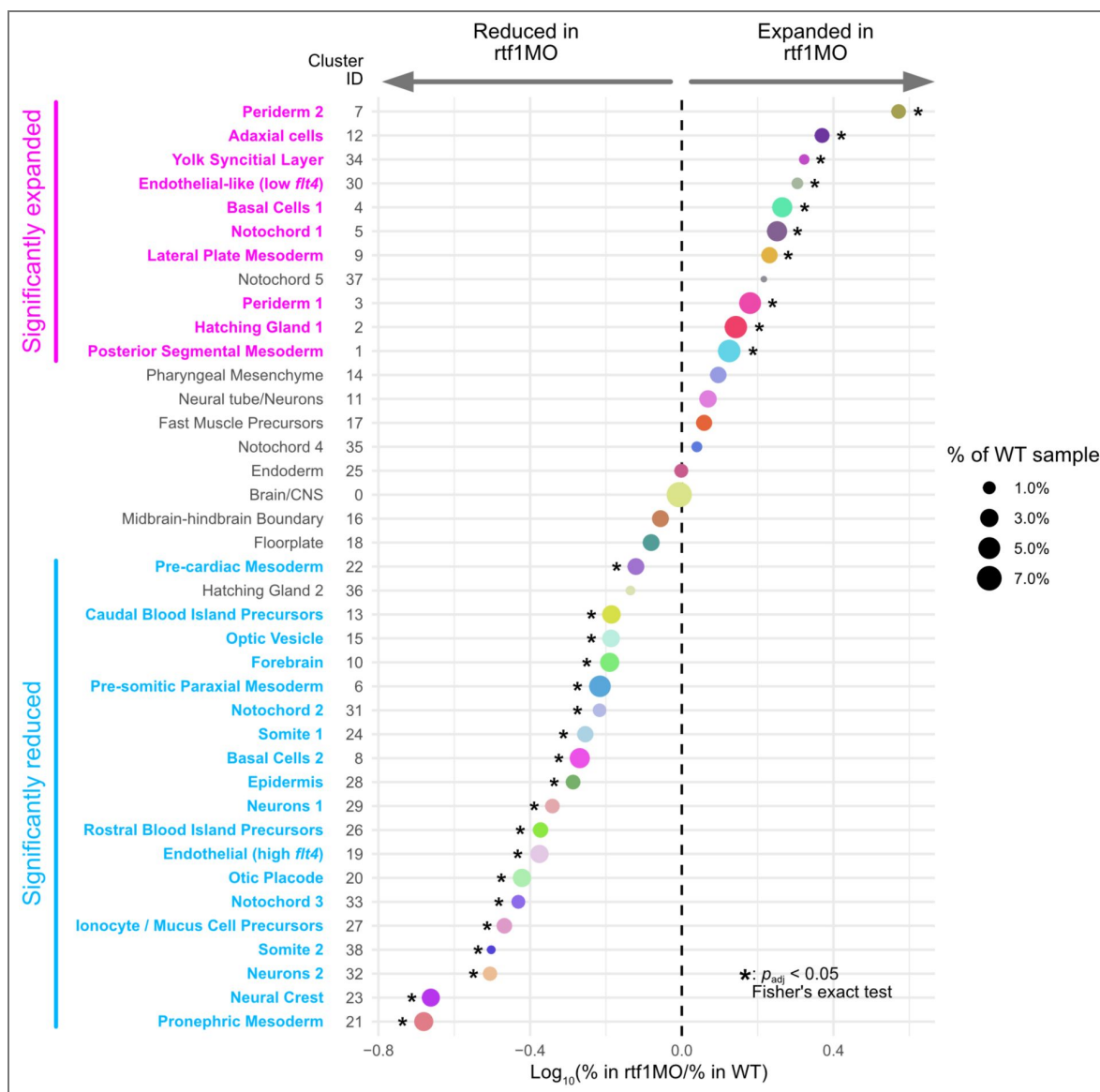


Figure 6 Supplement 3. Comparison of proportions of cells belonging to each Seurat cluster identified in the integrated multimodal single cell sequencing analysis of 11-12 somite stage control and *rtf1* morphant embryos.

Predicted identities of each cell type based on marker gene expression are displayed to the left of the Seurat cluster IDs. Cell type names are colored based on whether they are significantly expanded (pink) or significantly reduced (light blue) in *rtf1* morphants based on Fisher's exact test with an adjusted p -value cutoff of 0.05.

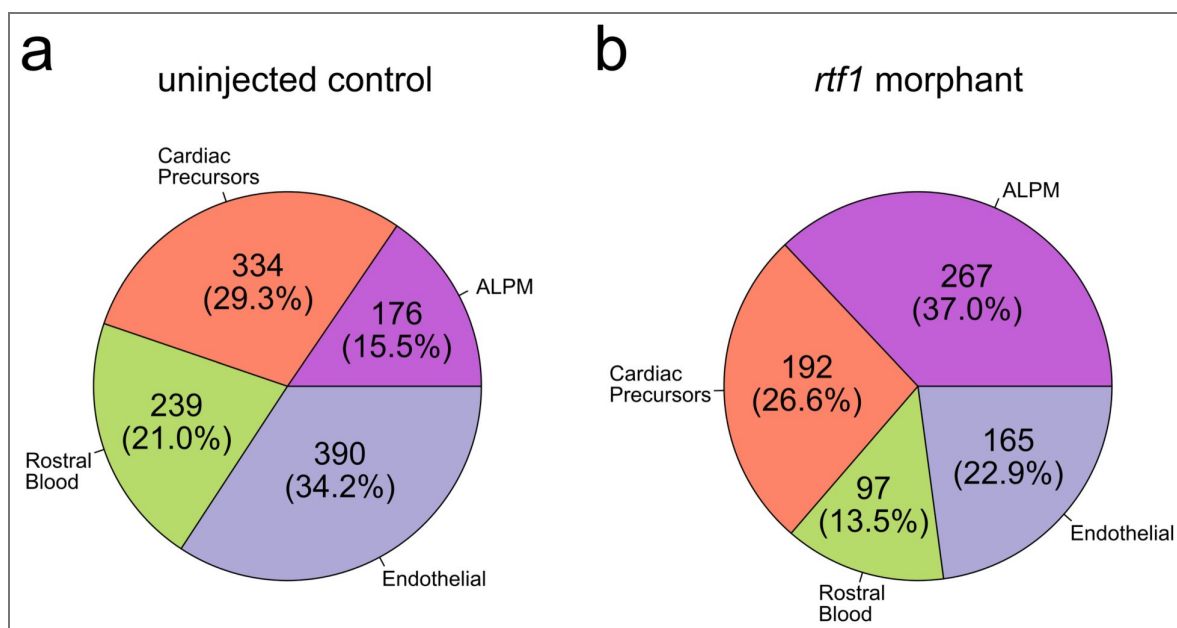


Figure 6 Supplement 4. Pie charts displaying the abundances of cell types identified in analysis of re-clustered anterior lateral plate mesoderm (ALPM) and derivatives from merged and integrated single cell RNA-seq datasets of 11-12 somite stage uninjected control (a) and *rtf1* morphant (b) embryos.

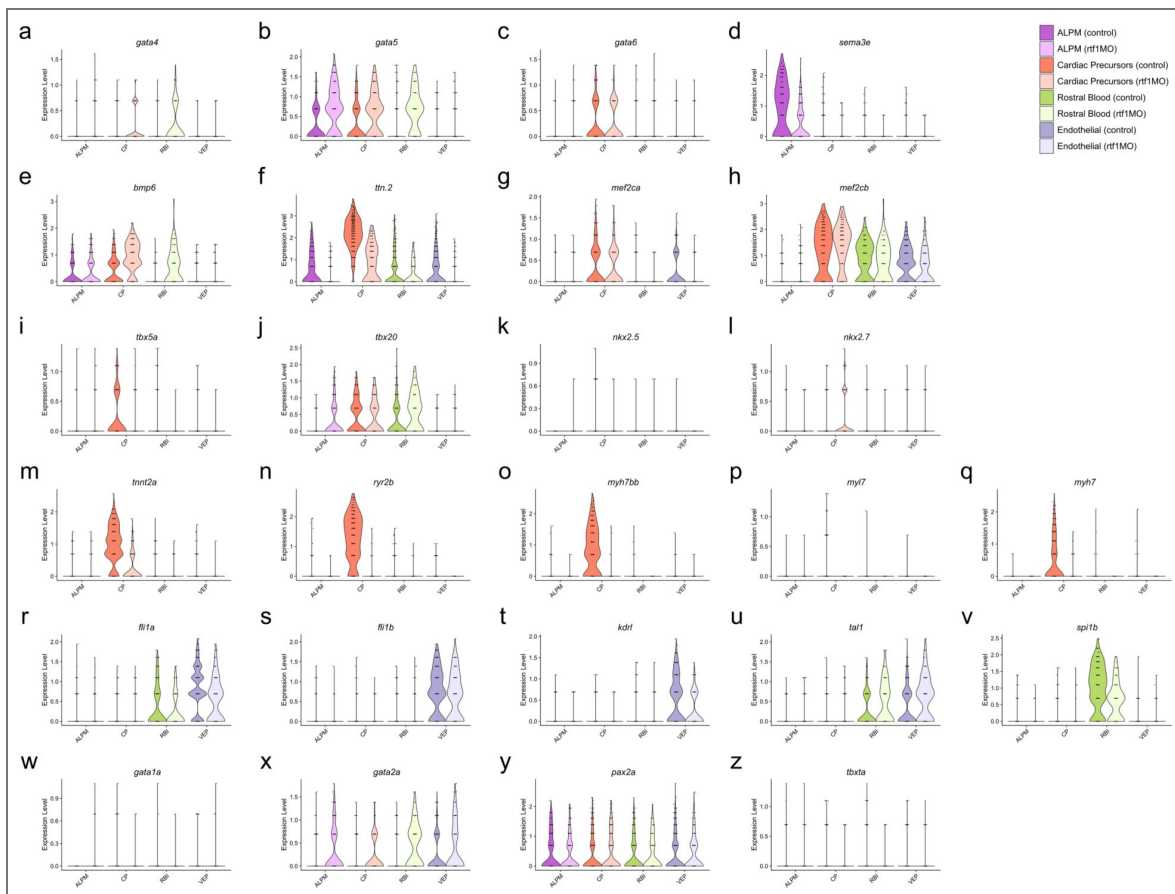


Figure 6 Supplement 5.

(a-z) Violin plots showing the expression of genes relevant to the development of the lateral plate mesoderm and its derivatives. Each dot represents the expression level of a gene in a single cell. Violin-shaped areas are color-coded based on cell types identified by re-clustering ALPM and derivative cells from merged and integrated single cell RNA-seq datasets of 11-12 somite stage uninjected control and *rtf1* morphant embryos. Shaded violins are not present when statistically significant expression of a particular gene was not detected.

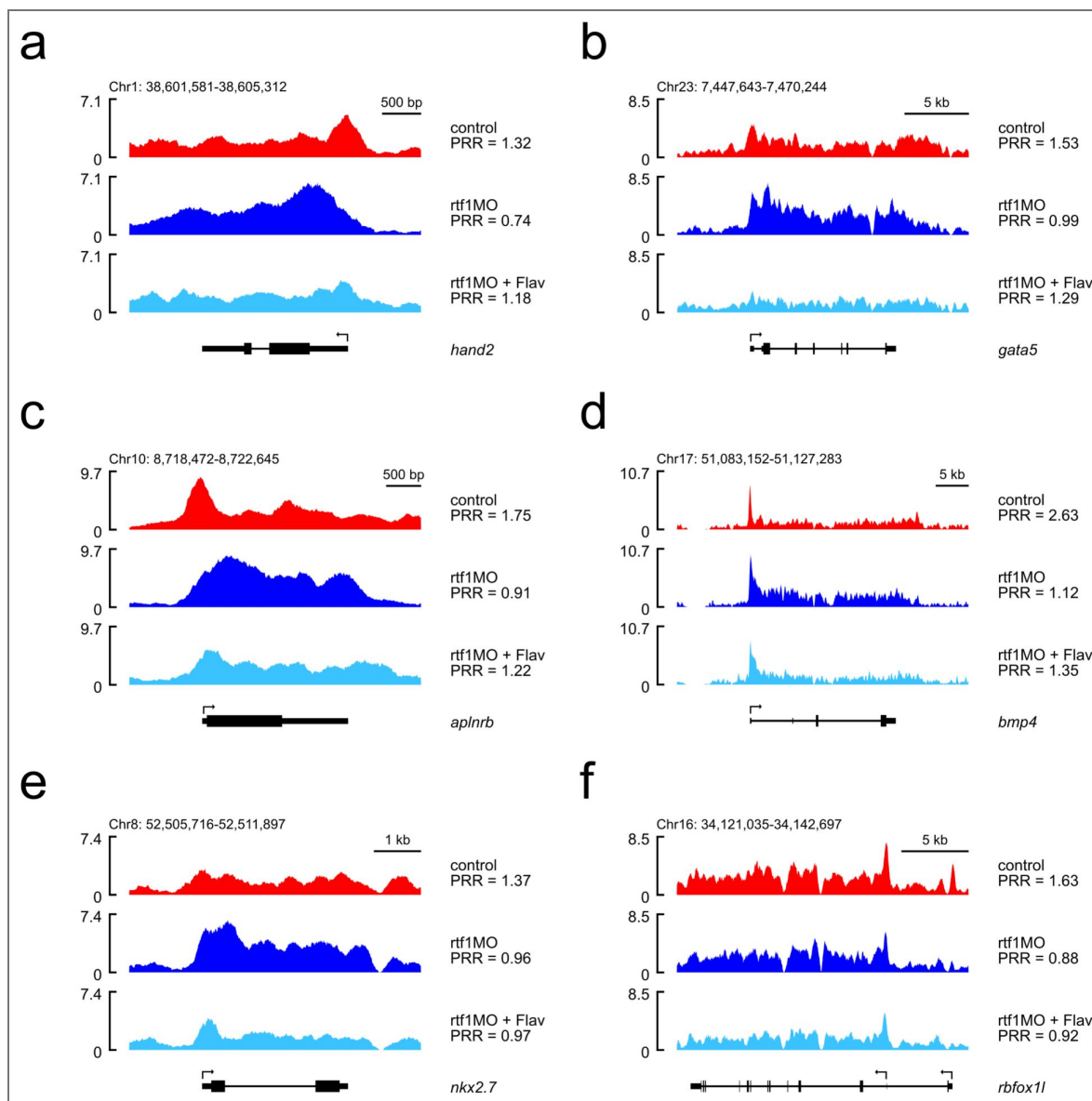


Figure 8 Supplement 1. Representative ChIP-seq tracks displaying RNA Pol II read densities (y-axis) for control (red), *rtf1* morphant (blue), and flavopiridol-treated *rtf1* morphant (light blue) embryos at cardiac mesoderm-related genes including *hand2* (a), *gata5* (b), *aplnrb* (c), *bmp4* (d), *nkx2.7* (e), and *rbfox11* (f).

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Additional information

Author Contributions

A.D.L.: study design, data acquisition, analysis, and interpretation, manuscript writing. F.L.: study design, data acquisition, analysis, and interpretation. L.T.: data acquisition. Z.Y.S.H.: data acquisition. B.K.: data acquisition. Z.T.: data acquisition. A.E.: data acquisition. M.P.: data analysis. H.N.: study design and data interpretation. A.N.: study design and data interpretation. J.-N.C.: study design, data interpretation, manuscript writing. All authors approved the final manuscript.

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Additional files

Supplementary Data File 1. [DESeq2](#) results file containing differential gene expression analysis data comparing FACS-isolated hand2:GFP-positive LPM from control and *rtf1* morphant 10-12 somite stage embryos.

Supplementary Data File 2. [Significantly restricted marker genes](#) for Seurat cell clusters from integrated multimodal (ATAC + Gene Expression) single cell sequencing dataset from control and *rtf1* morphant embryos at the 11-12 somite stage.

Supplementary Table 1. [Sequences of primers](#) used for quantitative real-time PCR analysis of gene expression in mouse embryonic stem cells.

Supplementary Table 2. [Predicted identities of cell types](#) belonging to each Seurat cluster identified in analysis of 11-12 somite stage control and *rtf1* morphant embryo integrated multimodal (ATAC + Gene Expression) single cell sequencing datasets. Predicted identities were based on manual inspection of marker gene expression.

References

- 1 Bruneau B. G (2013) Signaling and transcriptional networks in heart development and regeneration. *Cold Spring Harb Perspect Biol* **5**:a008292 <https://doi.org/10.1101/cshperspect.a008292>
- 2 Paige S. L., Plonowska K., Xu A., Wu S. M (2015) Molecular regulation of cardiomyocyte differentiation. *Circ Res* **116**:341-353 <https://doi.org/10.1161/CIRCRESAHA.116.302752>
- 3 Rana M. S., Christoffels V. M., Moorman A. F (2013) A molecular and genetic outline of cardiac morphogenesis. *Acta Physiol (Oxf)* **207**:588-615 <https://doi.org/10.1111/apha.12061>

- 4 Lu F., Langenbacher A. D., Chen J. N (2016) Transcriptional Regulation of Heart Development in Zebrafish. *J Cardiovasc Dev Dis* **3** <https://doi.org/10.3390/jcdd3020014>
- 5 Schoenebeck J. J., Keegan B. R., Yelon D (2007) Vessel and blood specification override cardiac potential in anterior mesoderm. *Dev Cell* **13**:254-267 <https://doi.org/10.1016/j.devcel.2007.05.012>
- 6 Mao L. M. F., Boyle Anderson E. A. T., Ho R. K (2021) Anterior lateral plate mesoderm gives rise to multiple tissues and requires tbx5a function in left-right asymmetry, migration dynamics, and cell specification of late-addition cardiac cells. *Dev Biol* **472**:52-66 <https://doi.org/10.1016/j.ydbio.2021.01.007>
- 7 Duong T. B., Holowiecki A., Waxman J. S (2021) Retinoic acid signaling restricts the size of the first heart field within the anterior lateral plate mesoderm. *Dev Biol* **473**:119-129 <https://doi.org/10.1016/j.ydbio.2021.02.005>
- 8 Sorrell M. R., Waxman J. S (2011) Restraint of Fgf8 signaling by retinoic acid signaling is required for proper heart and forelimb formation. *Dev Biol* **358**:44-55 <https://doi.org/10.1016/j.ydbio.2011.07.022>
- 9 Hota S. K., et al. (2019) Dynamic BAF chromatin remodeling complex subunit inclusion promotes temporally distinct gene expression programs in cardiogenesis. *Development* **146** <https://doi.org/10.1242/dev.174086>
- 10 Takeuchi J. K., et al. (2011) Chromatin remodelling complex dosage modulates transcription factor function in heart development. *Nat Commun* **2** <https://doi.org/10.1038/ncomms1187>
- 11 Lou X., Deshwar A. R., Crump J. G., Scott I. C (2011) Smarcd3b and Gata5 promote a cardiac progenitor fate in the zebrafish embryo. *Development* **138**:3113-3123 <https://doi.org/10.1242/dev.064279>
- 12 Lickert H., et al. (2004) Baf60c is essential for function of BAF chromatin remodelling complexes in heart development. *Nature* **432**:107-112 <https://doi.org/10.1038/nature03071>
- 13 Takeuchi J. K., Bruneau B. G (2009) Directed transdifferentiation of mouse mesoderm to heart tissue by defined factors. *Nature* **459**:708-711 <https://doi.org/10.1038/nature08039>
- 14 Endoh M., et al. (2004) Human Spt6 stimulates transcription elongation by RNA polymerase II in vitro. *Mol Cell Biol* **24**:3324-3336 <https://doi.org/10.1128/MCB.24.8.3324-3336.2004>
- 15 Hartzog G. A., Wada T., Handa H., Winston F (1998) Evidence that Spt4, Spt5, and Spt6 control transcription elongation by RNA polymerase II in *Saccharomyces cerevisiae*. *Genes Dev* **12**:357-369 <https://doi.org/10.1101/gad.12.3.357>
- 16 Keegan B. R., et al. (2002) The elongation factors Pandora/Spt6 and Foggy/Spt5 promote transcription in the zebrafish embryo. *Development* **129**:1623-1632 <https://doi.org/10.1242/dev.129.7.1623>
- 17 Wada T., et al. (1998) DSIF, a novel transcription elongation factor that regulates RNA polymerase II processivity, is composed of human Spt4 and Spt5 homologs. *Genes Dev* **12**:343-356 <https://doi.org/10.1101/gad.12.3.343>
- 18 Wier A. D., Mayekar M. K., Heroux A., Arndt K. M., VanDemark A. P (2013) Structural basis for Spt5-mediated recruitment of the Paf1 complex to chromatin. *Proc Natl Acad Sci U S A* **110**:17290-17295 <https://doi.org/10.1073/pnas.1314754110>
- 19 Chen J. J., et al. (2020) Spt5 Phosphorylation and the Rtf1 Plus3 Domain Promote Rtf1 Function through Distinct Mechanisms. *Mol Cell Biol* **40** <https://doi.org/10.1128/MCB.00150-20>
- 20 Hu S., et al. (2021) SPT5 stabilizes RNA polymerase II, orchestrates transcription cycles, and maintains the enhancer landscape. *Mol Cell* **81**:4425-4439.e4426, <https://doi.org/10.1016/j.molcel.2021.08.029>
- 21 Park J., Park S., Lee J. S (2023) Role of the Paf1 complex in the maintenance of stem cell pluripotency and development. *Febs J* **290**:951-961 <https://doi.org/10.1111/febs.16582>
- 22 Van Oss S. B., Cucinotta C. E., Arndt K. M (2017) Emerging Insights into the Roles of the Paf1 Complex in Gene Regulation. *Trends Biochem Sci* **42**:788-798 <https://doi.org/10.1016/j.tibs.2017.08.003>
- 23 Jaehning J. A (2010) The Paf1 complex: platform or player in RNA polymerase II transcription?. *Biochim Biophys Acta* **1799**:379-388 <https://doi.org/10.1016/j.bbagr.2010.01.001>

- 24 Langenbacher A. D., et al. (2011) The PAF1 complex differentially regulates cardiomyocyte specification. *Dev Biol* **353**:19-28 <https://doi.org/10.1016/j.ydbio.2011.02.011>
- 25 Nguyen C. T., Langenbacher A., Hsieh M., Chen J. N (2010) The PAF1 complex component Leo1 is essential for cardiac and neural crest development in zebrafish. *Dev Biol* **341**:167-175 <https://doi.org/10.1016/j.ydbio.2010.02.020>
- 26 Akanuma T., Koshida S., Kawamura A., Kishimoto Y., Takada S (2007) Paf1 complex homologues are required for Notch-regulated transcription during somite segmentation. *EMBO reports* **8**:858-863 <https://doi.org/10.1038/sj.embor.7401045>
- 27 Lu F., Langenbacher A., Chen J. N (2017) Tbx20 drives cardiac progenitor formation and cardiomyocyte proliferation in zebrafish. *Dev Biol* **421**:139-148 <https://doi.org/10.1016/j.ydbio.2016.12.009>
- 28 Langenbacher A. D., et al. (2023) Rtf1-dependent transcriptional pausing regulates cardiogenesis. *bioRxiv* <https://doi.org/10.1101/2023.10.13.562296>
- 29 Saga Y., et al. (1999) MesP1 is expressed in the heart precursor cells and required for the formation of a single heart tube. *Development* **126**:3437-3447 <https://doi.org/10.1242/dev.126.15.3437>
- 30 Warner M. H., Roinick K. L., Arndt K. M (2007) Rtf1 is a multifunctional component of the Paf1 complex that regulates gene expression by directing cotranscriptional histone modification. *Mol Cell Biol* **27**:6103-6115 <https://doi.org/10.1128/MCB.00772-07>
- 31 Vos S. M., Farnung L., Linden A., Urlaub H., Cramer P (2020) Structure of complete Pol II-DSIF-PAF-SPT6 transcription complex reveals RTF1 allosteric activation. *Nat Struct Mol Biol* **27**:668-677 <https://doi.org/10.1038/s41594-020-0437-1>
- 32 Cucinotta C. E., Hildreth A. E., McShane B. M., Shirra M. K., Arndt K. M (2019) The nucleosome acidic patch directly interacts with subunits of the Paf1 and FACT complexes and controls chromatin architecture in vivo. *Nucleic Acids Res* **47**:8410-8423 <https://doi.org/10.1093/nar/gkz549>
- 33 Van Oss S. B., et al. (2016) The Histone Modification Domain of Paf1 Complex Subunit Rtf1 Directly Stimulates H2B Ubiquitylation through an Interaction with Rad6. *Mol Cell* **64**:815-825 <https://doi.org/10.1016/j.molcel.2016.10.008>
- 34 Piro A. S., Mayekar M. K., Warner M. H., Davis C. P., Arndt K. M (2012) Small region of Rtf1 protein can substitute for complete Paf1 complex in facilitating global histone H2B ubiquitylation in yeast. *Proc Natl Acad Sci U S A* **109**:10837-10842 <https://doi.org/10.1073/pnas.1116994109>
- 35 de Jong R. N., et al. (2008) Structure and DNA binding of the human Rtf1 Plus3 domain. *Structure* **16**:149-159 <https://doi.org/10.1016/j.str.2007.10.018>
- 36 Core L., Adelman K (2019) Promoter-proximal pausing of RNA polymerase II: a nexus of gene regulation. *Genes Dev* **33**:960-982 <https://doi.org/10.1101/gad.325142.119>
- 37 Adelman K., Lis J. T (2012) Promoter-proximal pausing of RNA polymerase II: emerging roles in metazoans. *Nat Rev Genet* **13**:720-731 <https://doi.org/10.1038/nrg3293>
- 38 Peng J., Zhu Y., Milton J. T., Price D. H (1998) Identification of multiple cyclin subunits of human P-TEFb. *Genes Dev* **12**:755-762 <https://doi.org/10.1101/gad.12.5.755>
- 39 Price D. H (2000) P-TEFb, a cyclin-dependent kinase controlling elongation by RNA polymerase II. *Mol Cell Biol* **20**:2629-2634 <https://doi.org/10.1128/MCB.20.8.2629-2634.2000>
- 40 Paparidis N. F., Durvale M. C., Canduri F (2017) The emerging picture of CDK9/P-TEFb: more than 20 years of advances since PITALRE. *Mol Biosyst* **13**:246-276 <https://doi.org/10.1039/c6mb00387g>
- 41 Fujinaga K., Huang F., Peterlin B. M (2023) P-TEFb: The master regulator of transcription elongation. *Mol Cell* **83**:393-403 <https://doi.org/10.1016/j.molcel.2022.12.006>
- 42 Jonkers I., Kwak H., Lis J. T (2014) Genome-wide dynamics of Pol II elongation and its interplay with promoter proximal pausing, chromatin, and exons. *eLife* **3**:e02407 <https://doi.org/10.7554/eLife.02407>
- 43 Huang M., et al. (2022) Negative Elongation Factor (NELF) Inhibits Premature Granulocytic Development in Zebrafish. *Int J Mol Sci* **23** <https://doi.org/10.3390/ijms23073833>

- 44 Yang Q., et al. (2016) RNA polymerase II pausing modulates hematopoietic stem cell emergence in zebrafish. *Blood* **128**:1701-1710 <https://doi.org/10.1182/blood-2016-02-697847>
- 45 Gaertner B., Zeitlinger J (2014) RNA polymerase II pausing during development. *Development* **141**:1179-1183 <https://doi.org/10.1242/dev.088492>
- 46 Gilchrist D. A., et al. (2012) Regulating the regulators: the pervasive effects of Pol II pausing on stimulus-responsive gene networks. *Genes Dev* **26**:933-944 <https://doi.org/10.1101/gad.187781.112>
- 47 Adelman K., et al. (2009) Immediate mediators of the inflammatory response are poised for gene activation through RNA polymerase II stalling. *Proc Natl Acad Sci U S A* **106**:18207-18212 <https://doi.org/10.1073/pnas.0910177106>
- 48 Rougvie A. E., Lis J. T (1988) The RNA polymerase II molecule at the 5' end of the uninduced hsp70 gene of *D. melanogaster* is transcriptionally engaged. *Cell* **54**:795-804 [https://doi.org/10.1016/s0092-8674\(88\)91087-2](https://doi.org/10.1016/s0092-8674(88)91087-2)
- 49 Chen F. X., et al. (2017) PAF1 regulation of promoter-proximal pause release via enhancer activation. *Science* **357**:1294-1298 <https://doi.org/10.1126/science.aan3269>
- 50 Gilchrist D. A., et al. (2010) Pausing of RNA polymerase II disrupts DNA-specified nucleosome organization to enable precise gene regulation. *Cell* **143**:540-551 <https://doi.org/10.1016/j.cell.2010.10.004>
- 51 Gilchrist D. A., et al. (2008) NELF-mediated stalling of Pol II can enhance gene expression by blocking promoter-proximal nucleosome assembly. *Genes Dev* **22**:1921-1933 <https://doi.org/10.1101/gad.1643208>
- 52 Jurynek M. J., et al. (2019) The Paf1 complex and P-TEFb have reciprocal and antagonist roles in maintaining multipotent neural crest progenitors. *Development* **146** <https://doi.org/10.1242/dev.180133>
- 53 Wang Z., et al. (2022) Coordinated regulation of RNA polymerase II pausing and elongation progression by PAF1. *Sci Adv* **8**:eabm5504 <https://doi.org/10.1126/sciadv.abm5504>
- 54 Westerfield M. (2000) *The zebrafish book. A guide for the laboratory use of zebrafish (Danio rerio)* (4th ed.) Univ. of Oregon Press.
- 55 Farley F. W., Soriano P., Steffen L. S., Dymecki S. M (2000) Widespread recombinase expression using FLP_{ER} (flipper) mice. *Genesis* **28**:106-110
- 56 Chen J. N., Fishman M. C (1996) Zebrafish tinman homolog demarcates the heart field and initiates myocardial differentiation. *Development* **122**:3809-3816
- 57 Robu M. E., et al. (2007) p53 activation by knockdown technologies. *PLoS Genet* **3**:e78 <https://doi.org/10.1371/journal.pgen.0030078>
- 58 Yin C., Kikuchi K., Hochgreb T., Poss K. D., Stainier D. Y (2010) Hand2 regulates extracellular matrix remodeling essential for gut-looping morphogenesis in zebrafish. *Dev Cell* **18**:973-984 <https://doi.org/10.1016/j.devcel.2010.05.009>
- 59 Kim D., et al. (2013) TopHat2: accurate alignment of transcriptomes in the presence of insertions, deletions and gene fusions. *Genome Biol* **14** <https://doi.org/10.1186/gb-2013-14-4-r36>
- 60 Liao Y., Smyth G. K., Shi W (2014) featureCounts: an efficient general purpose program for assigning sequence reads to genomic features. *Bioinformatics* **30**:923-930 <https://doi.org/10.1093/bioinformatics/btt656>
- 61 Love M. I., Huber W., Anders S (2014) Moderated estimation of fold change and dispersion for RNA-seq data with DESeq2. *Genome Biol* **15** <https://doi.org/10.1186/s13059-014-0550-8>
- 62 Yu G., Wang L. G., Han Y., He Q. Y (2012) clusterProfiler: an R package for comparing biological themes among gene clusters. *Omics* **16**:284-287 <https://doi.org/10.1089/omi.2011.0118>
- 63 Chicas A., et al. (2010) Dissecting the unique role of the retinoblastoma tumor suppressor during cellular senescence. *Cancer Cell* **17**:376-387 <https://doi.org/10.1016/j.ccr.2010.01.023>

- 64 Langmead B., Salzberg S. L (2012) Fast gapped-read alignment with Bowtie 2. *Nat Methods* **9**:357-359 <https://doi.org/10.1038/nmeth.1923>
 - 65 Lun A. T., Smyth G (2016) K. csaw: a Bioconductor package for differential binding analysis of ChIP-seq data using sliding windows. *Nucleic Acids Res* **44**:e45 <https://doi.org/10.1093/nar/gkv1191>
 - 66 Lun A. T., Smyth G. K (2014) De novo detection of differentially bound regions for ChIP-seq data using peaks and windows: controlling error rates correctly. *Nucleic Acids Res* **42**:e95 <https://doi.org/10.1093/nar/gku351>
 - 67 Kurtenbach S., Harbour J. W. (2019) SparK: A Publication-quality NGS Visualization Tool. *bioRxiv* 845529 <https://doi.org/10.1101/845529>
 - 68 Lawson N. D., et al. (2020) An improved zebrafish transcriptome annotation for sensitive and comprehensive detection of cell type-specific genes. *eLife* **9** <https://doi.org/10.7554/eLife.55792>
 - 69 Hao Y., et al. (2021) Integrated analysis of multimodal single-cell data. *Cell* **184**:3573-3587.e3529, <https://doi.org/10.1016/j.cell.2021.04.048>
 - 70 Korsunsky I., et al. (2019) Fast, sensitive and accurate integration of single-cell data with Harmony. *Nat Methods* **16**:1289-1296 <https://doi.org/10.1038/s41592-019-0619-0>
 - 71 Song M., et al. (2022) GATA4/5/6 family transcription factors are conserved determinants of cardiac versus pharyngeal mesoderm fate. *Sci Adv* **8**:eabg0834 <https://doi.org/10.1126/sciadv.abg0834>
 - 72 Cao J., et al. (2019) The single-cell transcriptional landscape of mammalian organogenesis. *Nature* **566**:496-502 <https://doi.org/10.1038/s41586-019-0969-x>
- Langenbacher AD, Lu F, Tsang L, Huang ZYS, Keer B, Tian Z, Eide A, Pellegrini M, Nakano H, Nakano A, et al. (2023) Rtf1-dependent transcriptional pausing regulates cardiogenesis. NCBI BioProject. ID PRJNA1015262 <https://www.ncbi.nlm.nih.gov/bioproject/?term=PRJNA1015262>

Peer reviews

Reviewer #1 (Public review):

This is a highly original and impactful study that significantly advances our understanding of transcriptional regulation, in particular RNAPII pausing, during early heart development. The Chen lab has a long history of producing influential studies in cardiac morphogenesis, and this manuscript represents another thorough and mechanistically insightful contribution. The authors have thoroughly addressed this Reviewer's concerns and incorporated all of my suggestions in the revised manuscript. In addition, their responses to the other reviewer's comments are also very clear. As it is, this work is of great interest to the readership of Elife, as well as to the general scientific community.

The authors reveal a fundamentally new role for Rtf1-a component of the PAF1 complex-in governing promoter-proximal RNAPII pausing in the context of myocardial lineage specification. While transcriptional pausing has been implicated in stress responses and inducible gene programs, its developmental relevance has remained poorly defined. This study fills that gap with rigorous in vivo evidence demonstrating that Rtf1-dependent pausing is indispensable for activating the cardiac gene program from the lateral plate mesoderm.

Importantly, the study also provides compelling therapeutic implications. Showing that CDK9 inhibition-using either flavopiridol or targeted knockdown-can restore promoter-proximal pausing and rescue cardiomyocyte formation in Rtf1-deficient embryos suggests that modulation of pause-release kinetics may represent a new avenue for correcting transcriptionally driven congenital heart defects. Given that many CDK inhibitors are clinically approved or in active development, this connection significantly elevates the translational impact of the findings.

In sum, this study is rigorous, innovative, and transformative in its implications for developmental biology and cardiac medicine. I strongly support its publication.

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Reviewer #2 (Public review):

Summary:

Langenbacher et al. examine the requirement of Rtf1, a component of the PAF1C complex, which regulates transcriptional pausing in cardiac development. The authors first confirm that newly generated *rtf1* mutant alleles recapitulate the defects in cardiac progenitor differentiation found using morpholinos from their previous work. The authors then show that conditional loss of Rtf1 in mouse embryos and depletion in mouse ESCs both demonstrates a failure to turn on cardiac progenitor and differentiation marker genes, supporting conservation of Rtf1 in promoting vertebrate cardiac progenitor development. The authors then employ bulk RNA-seq on flow-sorted *hand2*:GFP⁺ cells and multiomic single-cell RNA-seq on whole Rtf1-depleted zebrafish embryos at the 10-12 somite stage. These experiments corroborate that gene expression associated with cardiac progenitor differentiation is lost. Furthermore, analysis of differentiation trajectories suggests that the expression of genes associated with cardiac, blood, and endothelial progenitor differentiation is not initiated within the anterior lateral plate mesoderm. Structure-function analysis supports that the Rtf1 Plus3 domain is necessary for its function in promoting cardiac progenitor differentiation. ChIP-seq for RNA Pol II on 10-12 somite stage zebrafish embryos supports that Rtf1 is required for proper promoter pausing at the transcriptional start site. The transcriptional promoter pausing defect and cardiac differentiation can partially be rescued in zebrafish *rtf1* mutants through pharmacological inhibition and depletion of Cdk9, a kinase that inhibits elongation. Thus, the authors have provided a clear analysis of the requirements and basic mechanism that Rtf1 employs regulating cardiac progenitor development.

Strengths and weaknesses:

Overall, the data presented are strong and the message of the study is clear. The conclusions that Rtf1 is required for transcriptional pause release and promotes vertebrate cardiac progenitor differentiation are supported. Areas of strength include the complementary approaches in zebrafish and mouse embryos, and mouse embryonic stem cells, which together support the conserved requirement for Rtf1 in promoting cardiac differentiation. The bulk and single-cell RNA-sequencing analyses provide further support for this model via examining broader gene expression. In particular, the pseudotime analysis bolsters that there is a broader effect on differentiation of anterior lateral plate mesoderm derivatives. The structure-function analysis provides a relatively clean demonstration of the requirement of the Rtf1 Plus3 domain. The pharmacological and depletion epistasis of Cdk9 combined with the RNA Pol II ChIP-seq nicely support the mechanism implicating Cdk9 in the Rtf1-dependent RNA Pol II promoter pausing. Additionally, this is a revised manuscript. The authors were overall responsive to the previous critiques. The new analysis and revisions have helped to strengthen their hypothesis and improve the clarity of their study. While the revised manuscript is significantly improved, the lack of analysis from the multiomic analysis still represents a lost opportunity to provide further insight into Rtf1 mechanisms within this study. However, the authors have nevertheless achieved their goal for this study. The data sets reported will also be useful tools for further analysis and integration by the cardiovascular development community. Thus, the study will be of interest to scientists studying cardiovascular development and those broadly interested in epigenetic regulation controlling vertebrate development.

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Author response:

The following is the authors' response to the original reviews.

Public Reviews:

Reviewer #1 (Public Review):

Summary:

The manuscript submitted by Langenbacher et al., entitled "Rtf1-dependent transcriptional pausing regulates cardiogenesis", describes very interesting and highly impactful observations about the function of Rtf1 in cardiac development. Over the last few years, the Chen lab has published novel insights into the genes involved in cardiac morphogenesis. Here, they used the mouse model, the zebrafish model, cellular assays, single cell transcription, chemical inhibition, and pathway analysis to provide a comprehensive view of Rtf1 in RNAPII (Pol2) transcription pausing during cardiac development. They also conducted knockdown-rescue experiments to dissect the functions of Rtf1 domains.

Strengths:

The most interesting discovery is the connection between Rtf1 and CDK9 in regulating Pol2 pausing as an essential step in normal heart development. The design and execution of these experiments also demonstrate a thorough approach to revealing a previously underappreciated role of Pol2 transcription pausing in cardiac development. This study also highlights the potential amelioration of related cardiac deficiencies using small molecule inhibitors against cyclin dependent kinases, many of which are already clinically approved, while many other specific inhibitors are at various preclinical stages of development for the treatment of other human diseases. Thus, this work is impactful and highly significant.

We thank the reviewer for appreciating our work.

Reviewer #2 (Public Review):

Summary:

Langenbacher et al. examine the requirement of Rtf1, a component of the PAF1C, which regulates transcriptional pausing in cardiac development. The authors first confirm their previous morphant study with newly generated rtf1 mutant alleles, which recapitulate the defects in cardiac progenitor and diUerentiation gene expression observed previously in morphants. They then examine the conservation of Rtf1 in mouse embryos and embryonic stem cell-derived cardiomyocytes. Conditional loss of Rtf1 in mesodermal lineages and depletion in murine ESCs demonstrates a failure to turn on cardiac progenitor and diUerentiation marker genes, supporting conservation of Rtf1 in promoting cardiac development. The authors subsequently employ bulk RNA-seq on flow-sorted hand2:GFP+ cells and multiomic single-cell RNA-seq on whole Rtf1-depleted embryos at the 10-12 stage. These experiments corroborate that genes associated with cardiac and muscle development are lost. Furthermore, the diUerentiation trajectories suggest that the expression of genes associated with cardiac maturation is not initiated. Structure-function analysis supports that the Plus3 domain is necessary for its function in promoting cardiac progenitor formation. ChIP-seq for RNA Pol II on 1012 somite stage embryos suggests that Rtf1 is required for proper promoter pausing. This defect can

partially be rescued through use of a pharmacological inhibitor for Cdk9, which inhibits elongation, can partially restore elongation in rtf1 mutants.

Strengths:

Many aspects of the data are strong, which support the basic conclusions of the authors that Rtf1 is required for transcriptional pausing and has a conserved requirement in vertebrate cardiac development. Areas of strength include the genetic data supporting the conserved requirement for Rtf1 in promoting cardiac development, the complementary bulk and single-cell RNA-sequencing approaches providing some insight into the gene expression changes of the cardiac progenitors, the structure-function analysis supporting the requirement of the Plus3 domain, and the pharmacological epistasis combined with the RNA Pol II ChIP-seq, supporting the mechanism implicating Cdk9 in the Rtf1 dependent mechanism of RNA Pol II pausing.

We thank the reviewer for the summary and for recognizing many strengths of our work.

Weaknesses:

While most of the basic conclusions are supported by the data, there are a number of analyses that are confusing as to why they chose to perform the experiments the way they did and some places where the interpretations presently do not support the interpretations. One of the conclusions is that the phenotype affects the maturation of the cardiomyocytes and they are arresting in an immature state. However, this seems to be mostly derived from picking a few candidates from the single cell data in Fig. 6. If that were the case, wouldn't the expectation be to observe relatively normal expression of earlier marker genes required for specification, such as Nkx2.5 and Gata5/6? The in situ expression analysis from fish and mice (Fig. 2 and Fig. 3) and bulk RNA-seq (Fig. 5) seems to suggest that there are pretty early specification and differentiation defects. While some genes associated with cardiac development are not changed, many of these are not specific to cardiomyocyte progenitors and expressed broadly throughout the ALPM. Similarly, it is not clear why a consistent set of cardiac progenitor genes (for instance mef2c, nkx2.5, and tbx20) was analyzed for all the experiments, in particular with the single cell analysis.

A major conclusion of our study is that Rtf1 deficiency impairs myocardial lineage differentiation from mesoderm, as suggested by the reviewer. Thus, the main goal of this study is to understand how Rtf1 drives cardiac differentiation from the LPM, rather than the maturation of cardiomyocytes. Multiple lines of evidence support this conclusion:

- (a) In situ hybridization showed that Rtf1 mutant embryos do not have nkx2.5+ cardiac progenitor cells and subsequently fail to produce cardiomyocytes (Figs. 2, 3).
- (b) RT-PCR analysis showed that knockdown of Rtf1 in mouse embryonic stem cells causes a dramatic reduction of cardiac gene expression and production of significantly fewer beating patches (Fig.4).
- (c) Bulk RNA sequencing revealed significant downregulation of cardiac lineage genes, including nkx2.5 (Fig. 5).
- (d) Single cell RNA sequencing clearly showed that lateral plate mesoderm (LPM) cells are significantly more abundant in Rtf1 morphants, whereas cardiac progenitors are less abundant (Fig. 6 and Fig.6 Supplement 1-5).

When feasible, we used cardiac lineage restricted markers in our assays. Nkx2.5 and tbx5a are not highlighted in the single cell analysis because their expression in our sc-seq dataset was too low to examine in the clustering/trajectory analysis. In this revised manuscript, we

provide violin plots showing the low expression levels of these genes in single cells from Rtf1 deficient embryos (Figure 6 Supplement 5).

The point of the multiomic analysis is confusing. RNA- and ATAC-seq were apparently done at the same time. Yet, the focus of the analysis that is presented is on a small part of the RNA-seq data. This data set could have been more thoroughly analyzed, particularly in light of how chromatin changes may be associated with the transcriptional pausing. This seems to be a lost opportunity. Additionally, how the single cell data is covered in Supplemental Fig. 2 and 3 is confusing. There is no indication of what the different clusters are in the Figure or the legend.

In this study, we performed single cell multiome analysis and used both scRNAseq and scATACseq datasets to generate reliable clustering. The scRNAseq analysis reveals how Rtf1 deficiency impacts cardiac differentiation from mesoderm, which inspired us to investigate the underlying mechanism and led to the discovery of defects in Rtf1-dependent transcriptional pause release.

We agree with the reviewer that deep examination of Rtf1-dependent chromatin changes would provide additional insights into how Rtf1 influences early development and careful examination of the scATACseq dataset is certainly a good future direction.

In this revised manuscript, we have revised Fig.6 Supplement 1 to include the predicted cell types and provide an additional excel file showing the annotation of all 39 clusters (Supplementary Table 2).

While the effect of Rtf1 loss on cardiomyocyte markers is certainly dramatic, it is not clear how well the mutant fish have been analyzed and how specific the effect is to this population. It is interpreted that the effects on cardiomyocytes are not due to "transfating" of other cell fates, yet supplemental Fig. 4 shows numerous effects on potentially adjacent cell populations. Minimally, additional data needs to be provided showing the live fish at these stages and marker analysis to support these statements. In some images, it is not clear the embryos are the same stage (one can see pigmentation in the eyes of controls that is not in the mutants/morphants), causing some concern about developmental delay in the mutants.

Single cell RNA sequencing showed an increased abundance of LPM cells and a reduced abundance of cardiac progenitors in Rtf1 morphants (Fig. 6 and Fig.6 Supplement 1-5). The reclustering of anterior lateral plate mesoderm (ALPM) cells and their derivatives further showed that cells representing undifferentiated ALPM were increased whereas cells representing all three ALPM derivatives were reduced. These findings indicate a defect in ALPM differentiation.

The reviewer questioned whether we examined stage-matched embryos. In our assay, Rtf1 mutant embryos were collected from crosses of Rtf1 heterozygotes. Each clutch from these crosses consists of ¼ embryos showing rtf1 mutant phenotypes and ¾ embryos showing wild type phenotypes which were used as control. Mutants and their wild type siblings were fixed or analyzed at the same time.

The reviewer questioned the specificity of the Rtf1 deficient cardiac phenotype and pointed out that Rtf1 mutant embryos do not have pigment cells around the eye. Rtf1 is a ubiquitously expressed transcriptional regulator. Previous studies in zebrafish have shown that Rtf1 deficiency significantly impacts embryonic development. Rtf1 deficiency causes severe defects in cardiac lineage and neural crest cell development; consequently, Rtf1 deficient embryos do not have cardiomyocytes and pigmentation (Langenbacher et al., 2011, Akanuma et al., 2007, and Jurynec et al., 2019). We now provide an image showing a 2-day-old Rtf1 mutant embryo and their wild type sibling to illustrate the cardiac, neural crest, and somitogenesis defects caused by loss of Rtf1 activity (Fig. 2 Supplement 1).

With respect to the transcriptional pausing defects in the Rtf1 deficient embryos, it is not clear from the data how this effect relates to the expression of the cardiac markers. This could have been directly analyzed with some additional sequencing, such as PRO-seq, which would provide a direct analysis of transcriptional elongation.

We showed that Rtf1 deficiency results in a nearly genome-wide decrease in promoterproximal pausing and downregulation of cardiac makers. Attenuating transcriptional pause release could restore cardiomyocyte formation in Rtf1 deficient embryos. In this revised manuscript, we provide additional RNAseq data showing that the expression levels of critical cardiac development genes such as *nkx2.5*, *tbx5a*, *tbx20*, *mef2ca*, *mef2cb*, *tnn.2*, and *ryr2b* are significantly rescued. We agree with the reviewer that further analyses using the PRO-seq approach could provide additional insights, but it is beyond the scope of this manuscript.

Some additional minor issues include the rationale that sequence conservation suggests an important requirement of a gene (line 137), which there are many examples this isn't the case, referencing figures panels out of order in Figs. 4, 7, and 8) as described in the text, and using the morphants for some experiments, such as the rescue, that could have been done in a blinded manner with the mutants.

We have clarified the rationale in this revised manuscript and made the eRort to reference figures in order.

The reviewer commented that rescue experiments “could have been done in a blinded manner with the mutants”. This was indeed how the flavopiridol rescue and *cdk9* knockdown experiments were carried out. Embryos from crosses of Rtf1 heterozygotes were collected, fixed after treatment and subjected to in situ hybridization. Embryos were then scored for cardiac phenotype and genotyped (Fig.8 d-g). Morpholino knockdown was used in genomic experiments because our characterization of *rtf1* morphants showed that they faithfully recapitulate the *rtf1* mutant phenotype during the timeframe of interest (Fig. 2).

Recommendations for the authors:

Reviewer #1 (Recommendations For The Authors):

This reviewer has a few suggestions below, aimed at improving the clarity and impact of the current study. Once these items are addressed, the manuscript should be of interest to the Elife reader.

Item 1. Strengthening the interaction between Rfh1 and CDK9 on Pol2 pausing.

The authors have convincingly shown that the chemical inhibition of CDK9 by flavopiridol can partially rescue the expression of cardiac genes in the zebrafish model. Although flavopiridol is FDA approved and has been a classical inhibitor for the dissection of CDK9 function, it also inhibits related CDKs (such as Flavopiridol (Alvocidib) competes with ATP to inhibit CDKs including CDK1, CDK2, CDK4, CDK6, and CDK9 with IC50 values in the 20-100 nM range) Therefore, this study could be more impactful if the authors can provide evidence on which of these CDKs may be most relevant during Rtf1-dependent cardiogenesis. To determine whether the observed cardiac defect indicates a preferential role for CDK9, or that other CDKs may also be able to provide partial rescue may be clarified using additional, more selective small molecules (e.g., BAY1251152, LDC000067 are commercially available).

The reviewer raised a reasonable concern about the specificity of flavopiridol. We thank the reviewer for the insightful suggestion and share the concern about specificity. To address this question, we have used an orthogonal testing through morpholino inhibition where we

directly targeted CDK9 and observed the same level of rescue, supporting a critical role of transcription pausing in cardiogenesis.

Item 2. Differences between CRISPR lines and morphants

Much of the work presented used Rtf1 morphants while the authors have already generated 2 CRISPR lines. What is the difference between morphants and mutants? The authors should comment on the similarities and/or differences between using morphants or mutants in their study and whether the same Rtf1- CDK9 connection also occurs in the CRISPR lines.

The morphology of our mutants (rtf1^{LA2678} and rtf1^{LA2679}) resembles the morphants and the previously reported ENU-induced rtf1^{KT641} allele. Extensive in situ hybridization analysis showed that the morphants faithfully recapitulate the mutant phenotypes (Fig.2). We have performed rescue experiments (flavopiridol and CDK9 morpholino) using Rtf1 mutant embryos and found that inhibiting Cdk9 restores cardiomyocyte formation (Fig.8).

Item 3. Discuss the therapeutic relevance of study

The authors have already generated a mouse model of Rtf1 Mesp1-Cre knockout where cardiac muscle development is severely derailed (Fig 3B). Thus, a demonstration of a conserved role for CDK9 inhibitor in rescuing cardiogenesis using mouse cells or the mouse model will provide important information on a conserved pathway function relevant to mammalian heart development. In the Discussion, how this underlying mechanistic role may be useful in the treatment of congenital heart disease should be provided.

Thank you for the insight. We have incorporated your comments in the discussion.

Item 4. Insights into the role of CDK9-Rtf1 in response to stress versus in cardiogenesis.

In the Discussion, the authors commented on the role of additional stress-related stimuli such as heat shock and inflammation that have been linked to CDK9 activity. However, the current ms provides the first, endogenous role of Pol2 pausing in a critical developmental step during normal cardiogenesis. The authors should emphasize the novelty and significance of their work by providing a paragraph on the state of knowledge on the molecular mechanisms governing cardiogenesis, then placing their discovery within this framework. This minor addition will also clarify the significance of this work to the broad readership of eLife.

Thank you for the suggestion. We have incorporated your comments and elaborate on the novelty and significance of our work in the discussion.

Reviewer #2 (Recommendations For The Authors):

(1) It is difficult to assess what the overt defects are in the embryos at any stages. Images of live images were not included in the supplement. Do these have a small, malformed heart tube later or are the embryos just deteriorating due to broad defects?

The Rtf1 deficient embryos do not produce nkx2.5+ cardiac progenitors. Consequently, we never observed a heart tube or detected cells expressing cardiomyocyte marker genes such as myl7. This finding is consistent with previous reports using rtf1 morphants and rtf1^{KT64}, an ENU-induced point mutation allele (Langenbacher et al., 2011 and Akanuma, 2007). In this revised manuscript, we provide a live image of 2-day-old wild type and rtf1^{LA2679/LA2679} embryos (Fig. 2 Supplement 1). After two days, rtf1 mutant embryos undergo broad cell death.

(2) Fig. 2, although the *in situs* are convincing, there is not a quantitative assessment of expression changes for these genes. This could have been done for the bulk or single cell RNA-seq experiments, but was not and these genes weren't not included in the heat maps. A quantitative assessment of these genes would benefit the study.

The top 40 most significantly differentially expressed genes are displayed in the heatmap presented in Fig.5d. The complete differential gene expression analysis results for our hand2 FACS-based comparison of *rtf1* morphants and controls is presented in Supplementary Data File 1. In this revised manuscript, we provide a new supplemental figure with violin plots showing the expression levels of genes of interest in our single cell sequencing dataset (Fig.6 Supplement 5).

(3) It doesn't not appear that any statistical tests were used for the comparisons in Fig. 2.

We now provide the statistical data in the legend and Fig.2 b, d, f, h and i.

(4) It's not clear the magnifications and orientations of the embryos in Fig. 3b are the same.

Embryos shown in Fig.3b are at the same magnification. However, because *Rtf1* mutant embryos display severe morphological defects, the orientation of mutant embryos was adjusted to examine the cardiac tissue.

(5) The *n*'s for analysis of *MLC2v* in WT *Rtf1* CKO embryos in Fig. 3b are only 1. At least a few more embryos should be analyzed to confirm that the phenotype is consistent.

We have revised the figure and present the number of embryos analyzed and statistics in Fig.3c.

(6) A number of figure panels are referred to out of order in the text. Fig. 4E-G are before Fig. 4C, D, Fig. 7C before 7B, Fig. 8D-I before 8A, B. In general, it is easier for the reader if the figures panels are presented in the order they are referred to in the text.

Revised as suggested.

(7) While additional genes can be included, it is not clear why the same sets of genes are not examined in the bulk or single-cell RNA-seq as with the *in situs* or expression was analyzed in embryos. I suggest including the genes like *nkx2.5*, *tbx20*, *myl7*, in all the sequencing analysis.

We used the same set of genes in all analyses when possible. However, the low expression of genes such as *nkx2.5* and *myl7* in our sc-seq dataset preclude them from the clustering/trajectory analysis. In this revised manuscript, we present violin plots showing their expression in wild type and *rtf1* morphants (Fig. 6 Supplement 5).

(8) If a multiomic approach was used, why wasn't its analysis incorporated more into the manuscript? In general, a clearer presentation and deeper analysis of the single cell data would benefit the study. The integration of the RNA and ATAC would benefit the analysis.

As addressed in our response to the reviewer's public review, both datasets were used in clustering. Examining changes in chromatin accessibility is certainly interesting, but beyond the scope of this study.

(9) Many of the markers analyzed are not cardiac specific or it is not clear they are expressed in cardiac progenitors at the stage of the analysis. *Hand2* has broader expression. Additional confirmation of some of the genes through *in situ* would help the interpretations.

Markers used for the in situ hybridization analysis (myl7, mef2ca, nkx2.5, tbx5a, and tbx20) are known for their critical role in heart development. For sc-seq trajectory analyses, most displayed genes (sema3e, bmp6, ttn.2, mef2cb, tnnt2a, ryr2b, and myh7bb) were identified based on their differential expression along the LPM-cardiac progenitor pseudotime trajectory. Rather than selecting genes based on their cardiac specificity, our goal was to examine the progressive gene expression changes associated with cardiac progenitor formation and compare gene expression of wild type and rtf1 deficient embryos.

| (10) Additional labels of the cell clusters are needed for Supplemental Figs. 2 and 3.

The cluster IDs were presented on Supplementary Figures 2 and 3. In this revised version, we added predicted cell types to the UMAP (revised Fig.6 Supplement 1) and provided an excel file with this information (revised Supplementary Table 2).

| (11) On lines 101-102, the interpretation from the previous data is that differentiation of the LPM requires Rtf1. However, later from the single cell data the interpretation based on the markers is that Rtf1 loss affects maturation. However, it is not clear this interpretation is correct or what changed from the single cell data. If that were the case, one would expect to see maintenance of more early marks and subsequent loss of maturation markers, which does not appear to be the case from the presented data.

Our data suggests that cardiac progenitor formation is not accomplished by simultaneously switching on all cardiac marker genes. Our pseudotime trajectory analysis highlights tnnt2a, ryr2b, and myh7bb as genes that increase in expression in a lagged manner compared to mef2cb (Fig. 6). Thus, the abnormal activation of mef2cb without subsequent upregulation of tnnt2a, ryr2b, and myh7bb in rtf1 morphants suggests a requirement for rtf1 in the progressive gene expression changes required for proper cardiac progenitor differentiation. Our single cell experiment focuses on the process of cardiac progenitor differentiation and does not provide insights into cardiomyocyte maturation. We have edited the text to clarify these interpretations.

| (12) The interpretation that there is not "transfating" is not supported by the shown data. Analysis of markers in other tissues, again with in situ, to show spatially would benefit the study.

As stated in our response to the reviewer's public review, we observed a dramatic increase of ALPM cells, but a decrease of ALPM derivatives including the cardiac lineage. We did not observe the expansion of one ALPM-derived subpopulation at the expense of the others. These observations suggest a defect in ALPM differentiation and argue against the notion that the region of the ALPM that would normally give rise to cardiac progenitors is instead differentiating into another cell type.

| (13) The rationale that sequence conservation means a gene is important (lines 137-139) is not really true. There are examples a lot of highly conserved genes whose mutants don't have defects.

We have revised the text to avoid confusion.

| (14) The data showing that the 8 bp mutations do not affect the RNA transcript is not shown or at least indicated in Fig. 7. It would seem that this experiment could have been done in the mutant embryos, in which case the experiment would have been semi-blinded as the genotyping would occur after imaging.

The modified Rtf1 wt RNA (Rtf1 wt* in revised Fig. 7) robustly rescued nkx2.5 expression in rtf1 deficient embryos, demonstrating that the 8 bp modifications do not negatively impact the activity of the injected RNA. As stated previously, morpholino knockdown was used in

some experiments because our characterization of rtf1 morphants showed that they faithfully recapitulate the rtf1 mutant phenotype during the timeframe of interest.

(15) Using a technique like PRO-seq at the same stage as the ChIP-seq would complement the ChIP-seq and allow a more detailed analysis of the transcriptional pausing on specific genes observed in WT and mutant embryos.

As stated in our response to the reviewer's public review, we appreciate the suggestion but PRO-seq is beyond the scope of this study.

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